CHARGEPULSE EV

GENAI PRACTICAL ANALYTICS ASSIGNMENT

Data Quality • Feature Engineering • Operations Analytics • Executive Dashboard

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Dataset: ChargePulse_EV_Operations

Prepared using the provided assignment brief and dataset. Metrics, assumptions and transformations are documented question-wise.

Methodology & Cleaning Notes

The provided workbook contained 2,232 raw records. After exact-row deduplication (18) and keeping the first record for repeated Session_ID values (0), the analytical table contains 2,214 records.

Validation assumptions used throughout this report

• Energy_kWh: valid when > 0 and ≤ 180. • Duration_Min: valid when > 0 and ≤ 720. • Customer_Rating: valid from 1 to 5. • Discount_Pct: 0–100. • Downtime_Min: 0–1,440. Invalid numeric values are marked missing rather than replaced with invented values. Session_Date_Raw is parsed day-first; Session_Start is used as a fallback where missing. Obvious category spelling/case variants are standardized.

Interpretation note

The composite scores and forecast are analytical models built for this assignment. Their assumptions and weights are stated in the relevant questions. Observed relationships do not prove causation. The prompts below are the exact prompts used to define each analysis in this report. This PDF contains generated analysis outputs and charts, not screenshots of a separate ChatGPT conversation; attach genuine screenshots separately if your instructor specifically requires UI screenshots.

Q1. GenAI-Assisted Data Quality Audit & Cleaning Plan

Problem Statement

Audit missing values, duplicates, category inconsistencies, numerical validity, and date formatting before analysis.

GenAI Prompt

Act as a senior data-quality analyst. Audit the EV_Charging_Raw_Data sheet. Report missing values by column, exact duplicate rows, duplicate Session_ID values, inconsistent City/Payment_Method/Session_Status labels, invalid Energy_kWh (<=0 or >180), Duration_Min (<=0 or >720), Customer_Rating (outside 1–5), and mixed Session_Date_Raw formats. Do not invent values. Recommend transparent cleaning actions and preserve raw columns.

Analysis / Outcome

Raw rows: 2,232. Exact duplicate rows removed: 18. Duplicate Session_ID rows removed after keeping the first occurrence: 0. Final analytical rows: 2,214. Invalid numeric values were set to missing using the stated validation rules; dates were parsed with day-first handling and Session_Start used as fallback when needed. Category labels were standardized for obvious spelling/case variants. Highest missing fields: Session_Date (70), Customer_Rating (49), Charging_Speed_kWh_per_Hour (46), Energy_kWh (38), Revenue_per_kWh (38), Zone (35).

Key Insights & Recommended Action

Cleaning rules are explicit and reproducible. Missing numeric values are not silently fabricated; metrics use available valid observations. Duplicate Session_ID handling may discard later records, so in a production workflow it should be confirmed with the data owner.

Supporting Results Table

Column	Missing_Count	Missing_%
Session_Date	70.00	3.16
Customer_Rating	49.00	2.21
Charging_Speed_kWh_per_Hour	46.00	2.08
Energy_kWh	38.00	1.72
Revenue_per_kWh	38.00	1.72
Zone	35.00	1.58
Connector_Type	31.00	1.40
Technician	30.00	1.36
Payment_Method	24.00	1.08
City	22.00	0.99
Duration_Min	8.00	0.36

Q2. GenAI-Assisted Feature Engineering for EV Charging Analysis

Problem Statement

Create five analytical features while preserving the original raw fields.

GenAI Prompt

Using Energy_kWh, Duration_Min, Final_Amount, Gross_Amount and Session_Start, define Charging_Speed_kWh_per_Hour = Energy_kWh/(Duration_Min/60), Revenue_per_kWh = Final_Amount/Energy_kWh, Discount_Amount = max(Gross_Amount-Final_Amount,0), Session_Hour = hour(Session_Start), and Time_Band using Night 00–05, Morning 06–11, Afternoon 12–17, Evening 18–23. Validate division denominators and leave unavailable results missing.

Analysis / Outcome

Added Charging_Speed_kWh_per_Hour, Revenue_per_kWh, Discount_Amount, Session_Hour and Time_Band. Speed is calculated only when energy and duration are valid and duration is positive. Revenue per kWh is undefined when energy is zero/missing. Discount_Amount is calculated from gross minus final amount and clipped at zero.

Key Insights & Recommended Action

These features support charger efficiency, revenue realization, discount cost and time-of-day demand. Extreme speeds should be investigated as outliers rather than interpreted automatically as superior performance.

Supporting Results Table

Feature	Logic
Charging_Speed_kWh_per_Hour	Energy_kWh ÷ (Duration_Min/6
Revenue_per_kWh	Final_Amount ÷ Energy_kWh
Discount_Amount	max(Gross_Amount – Final_Amo
Session_Hour	Hour from Session_Start
Time_Band	Night/Morning/Afternoon/Even

Q3. AI-Assisted City-Level Charging Network Performance Analysis

Problem Statement

Compare cities using volume, energy, revenue, completion/failure rates, downtime and customer rating.

GenAI Prompt

Analyze city performance after standardizing City and Session_Status. For each city calculate sessions, energy, final revenue, completion rate, failure rate, average downtime and average rating. Identify high-performing and high-attention cities without relying only on volume.

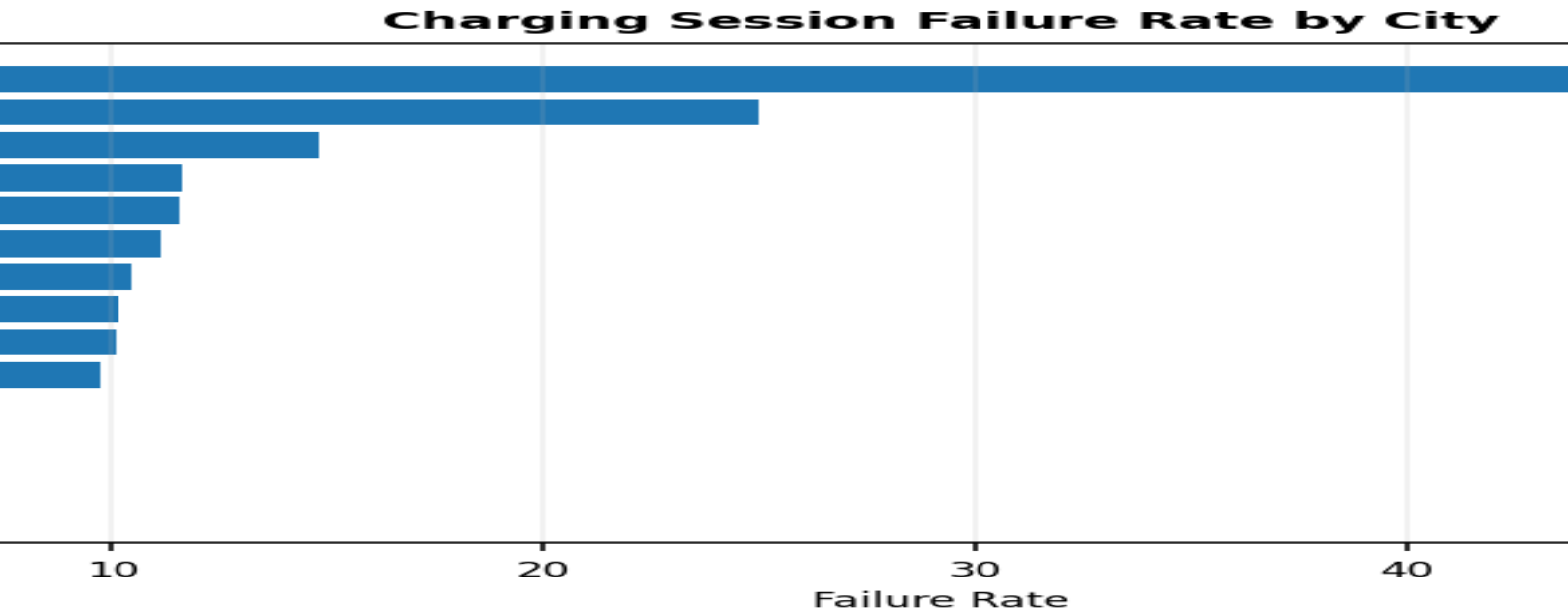
Analysis / Outcome

Highest observed failure-rate city: delhi ncr (55.6%). Strongest city by completion rate with rating as tie-break: bengaluru (88.9% completion; 4.11/5 average rating). City-level results are shown below.

Key Insights & Recommended Action

Prioritize the high-failure city for charger diagnostics and field checks. Validate sample sizes before treating small-city differences as stable.

Visualization



Supporting Results Table

City	Total_Sessions	Total_Energy	Total_Revenue	Completion_Rate	Failure_Rate	Avg_Downtime	Avg_Rating
delhi ncr	9.00	101.15	1,712.33	44.44	55.56	35.90	2.56
Delhi-NCR	4.00	154.86	2,602.70	50.00	25.00	12.00	3.25
Hyderabad	297.00	10,160.12	173,711.89	69.02	14.81	12.30	3.09
Chennai	249.00	8,951.51	152,974.06	70.28	11.65	10.53	3.18
Ahmedabad	216.00	7,872.80	137,153.28	67.59	11.57	12.02	3.14
bengaluru	9.00	260.76	3,684.46	88.89	11.11	1.99	4.11
Bangalore	230.00	5,327.77	90,943.22	72.61	10.43	10.46	3.16
Delhi	177.00	5,125.56	89,955.74	71.75	10.17	11.28	3.24
Pune	326.00	11,920.21	205,292.39	70.55	10.12	10.04	3.24
Mumbai	402.00	13,660.22	240,410.31	73.38	9.70	10.24	3.29
Jaipur	262.00	10,433.38	176,517.52	77.48	7.25	8.94	3.27
BENGALURU	4.00	193.92	3,003.23	75.00	0.00	7.35	3.50

Q4. GenAI-Assisted Charger Efficiency Analysis

Problem Statement

Compare charger types for energy/session, duration, speed, completion/failure and downtime.

GenAI Prompt

Group valid sessions by Charger_Type. Calculate total sessions, average energy, average duration, average charging speed, completion rate, failure rate and average downtime. Identify fastest and most reliable charger types separately; do not equate speed alone with efficiency.

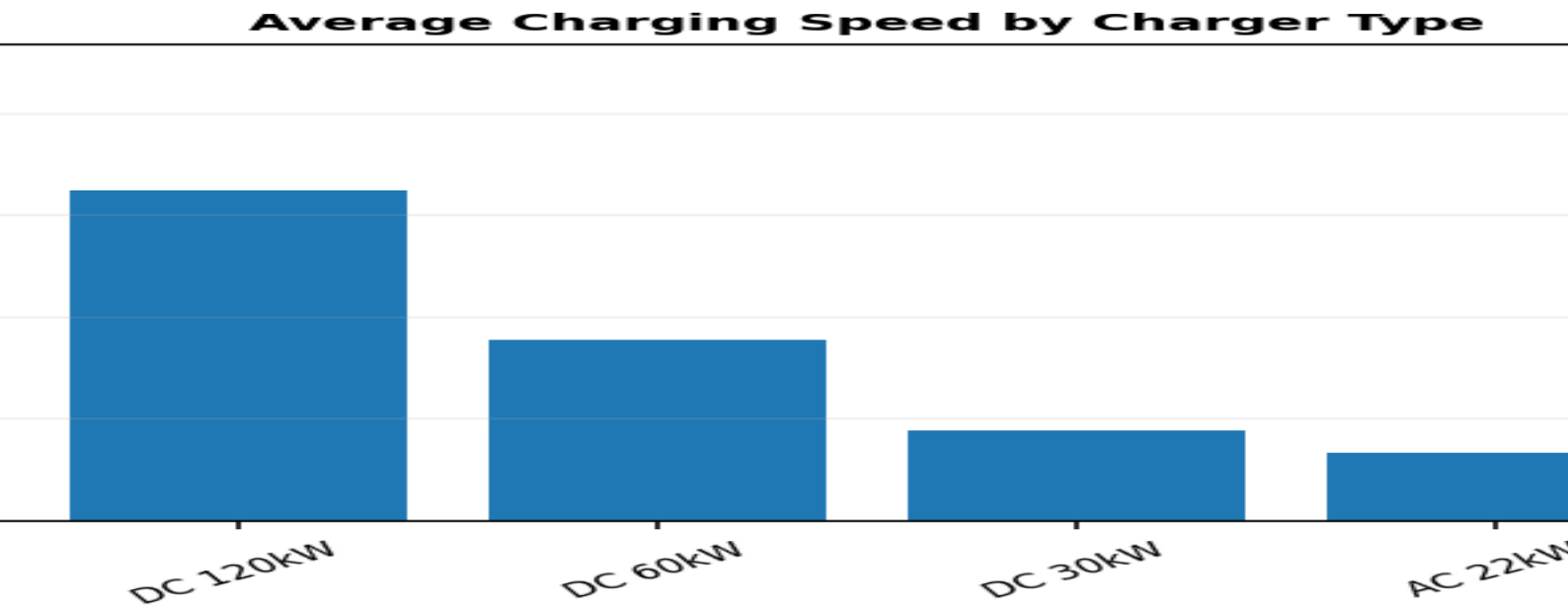
Analysis / Outcome

Highest average calculated charging speed: DC 180kW (89.19 kWh/hour). Highest completion rate: AC 7.4kW (76.2%). Compare the table to detect any speed-versus-reliability mismatch.

Key Insights & Recommended Action

Fast charging is useful only when accompanied by reliability. Review charger maintenance and downtime before expansion decisions.

Visualization



Supporting Results Table

Charger_Type	Total_Sessions	Avg_Energy	Avg_Speed	Completion_Rate	Failure_Rate	Avg_Downtime
DC 180kW	290.00	79.90	89.19	69.66	10.00	10.02
DC 120kW	371.00	64.34	64.92	70.35	12.67	10.62
DC 60kW	358.00	37.89	35.67	66.48	10.89	13.56
DC 30kW	374.00	20.86	17.87	74.33	11.50	10.24
AC 22kW	388.00	14.51	13.41	71.39	11.34	10.78
AC 7.4kW	433.00	5.12	4.61	76.21	8.31	9.77

Q5. GenAI-Assisted Payment Method & Revenue Analysis

Problem Statement

Compare payment channel volume, gross/final revenue, discounts and reliability.

GenAI Prompt

Standardize Payment_Method and Session_Status. For each payment method calculate sessions, gross revenue, final revenue, average transaction value, total discount, completion rate and failure rate. Identify the largest final-revenue contributor and channels needing operational attention.

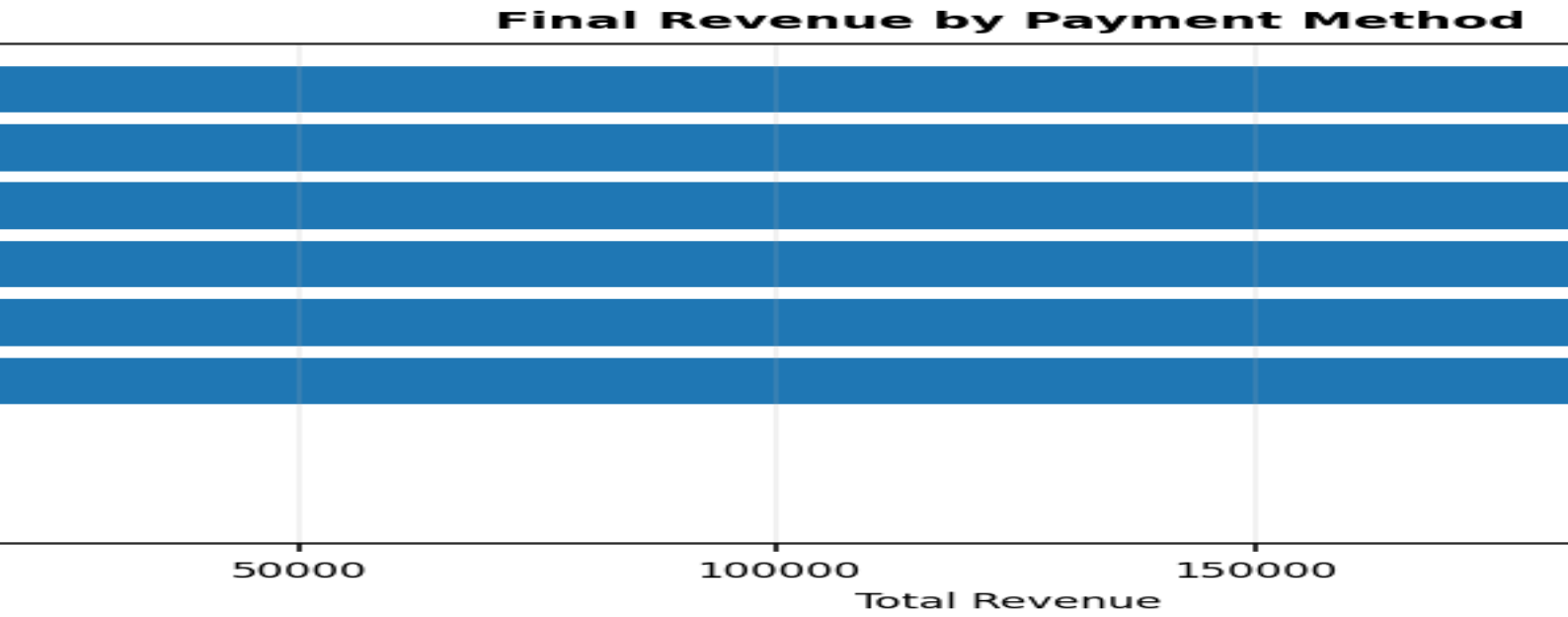
Analysis / Outcome

Largest final-revenue contributor: Credit Card (■236,559.62). Payment channel reliability and discount costs are shown in the table; revenue alone should not determine channel quality.

Key Insights & Recommended Action

Investigate payment methods with above-average failure rates for transaction retries, reconciliation issues and user-experience friction.

Visualization



Supporting Results Table

Payment_Method	Total_Sessions	Total_Revenue	Avg_Transaction	Total_Discount	Completion_Rate	Failure_Rate
Credit Card	396.00	236,559.62	597.37	12,027.74	66.92	12.12
UPI	373.00	223,422.84	598.99	11,498.80	72.92	8.31
Corporate Account	350.00	222,060.71	634.46	12,917.94	73.14	10.57
App Wallet	352.00	206,202.22	585.80	9,966.69	75.28	11.36
Debit Card	364.00	204,596.91	562.08	9,152.93	73.08	10.16
RFID Wallet	349.00	184,676.35	529.16	10,249.43	69.91	11.75
Unknown	24.00	15,938.81	664.12	842.35	62.50	12.50
CreditCard	6.00	2,741.54	456.92	0.00	50.00	16.67

Q6. GenAI-Powered Customer Issue Classification from Unstructured Text

Problem Statement

Classify Issue_Notes by meaning and compare category frequency, problem sessions, ratings and downtime.

GenAI Prompt

Classify each Issue_Notes entry by semantic meaning into a practical issue category (payment/billing, connector/cable, app/connectivity, power supply, charging interruption/speed, other, missing). Use Session_Status, Customer_Rating and Downtime_Min to calculate counts, failed/interrupted sessions, problem-session rate, average rating and downtime. Treat this as rule-assisted classification requiring manual QA.

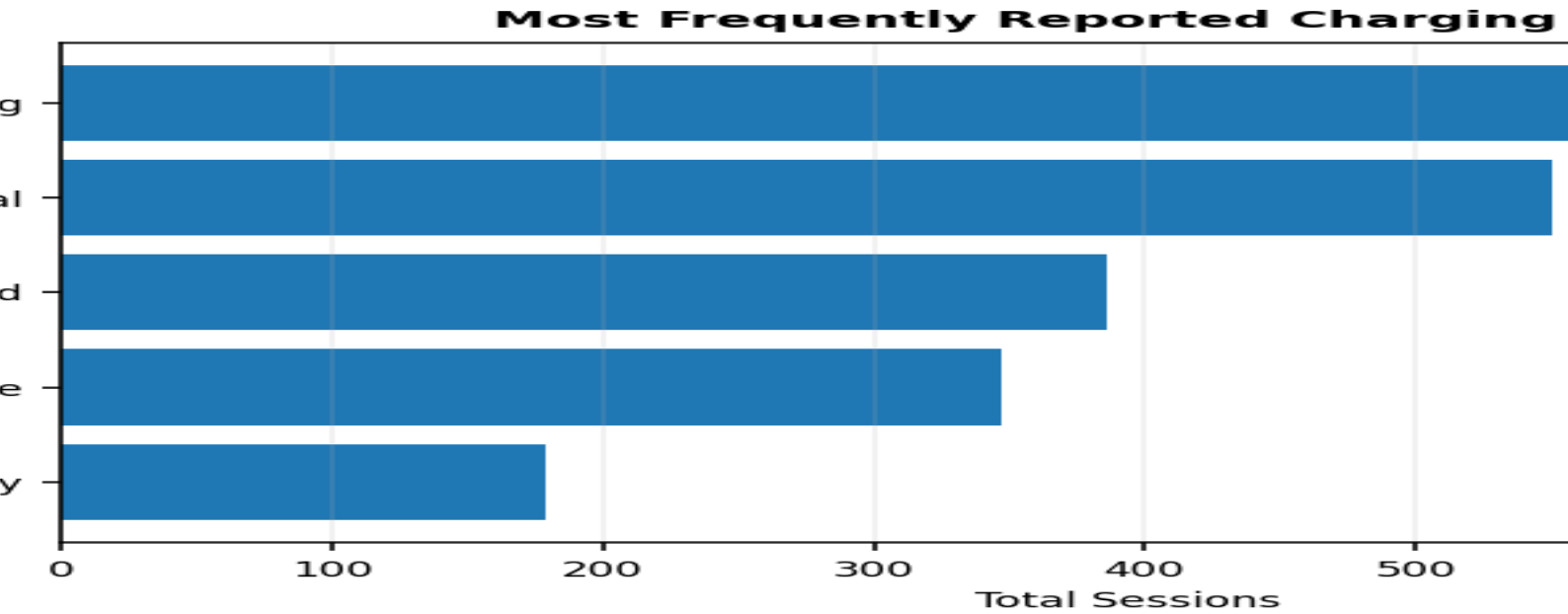
Analysis / Outcome

Most frequent category: Payment / Billing (751 records). Category with highest problem rate in this dataset: Connector / Cable (29.4%). Categories are assigned using transparent keyword/meaning rules, not an external LLM classification API; ambiguous notes may need manual review.

Key Insights & Recommended Action

Use the categories to prioritize recurring issues, but audit a sample of notes before using this classification for formal operational reporting.

Visualization



Supporting Results Table

Issue_Category	Total_Sessions	Problem_Rate	Avg_Downtime	Avg_Rating
Payment / Billing	751.00	28.89	10.46	3.23
Other / General	551.00	27.04	11.32	3.20
Charging Interruption / Spee	386.00	28.24	11.08	3.25
Connector / Cable	347.00	29.39	10.17	3.16
App / Connectivity	179.00	28.49	11.45	3.17

Q7. GenAI-Assisted Weather Impact Analysis on Charging Reliability

Problem Statement

Compare completion, failure, interruption, downtime, ratings and energy across weather conditions.

GenAI Prompt

Group sessions by Weather and calculate total sessions, completion rate, failure rate, interruption rate, average downtime, customer rating and energy. Compare failure and interruption separately and avoid claiming causation from observational data.

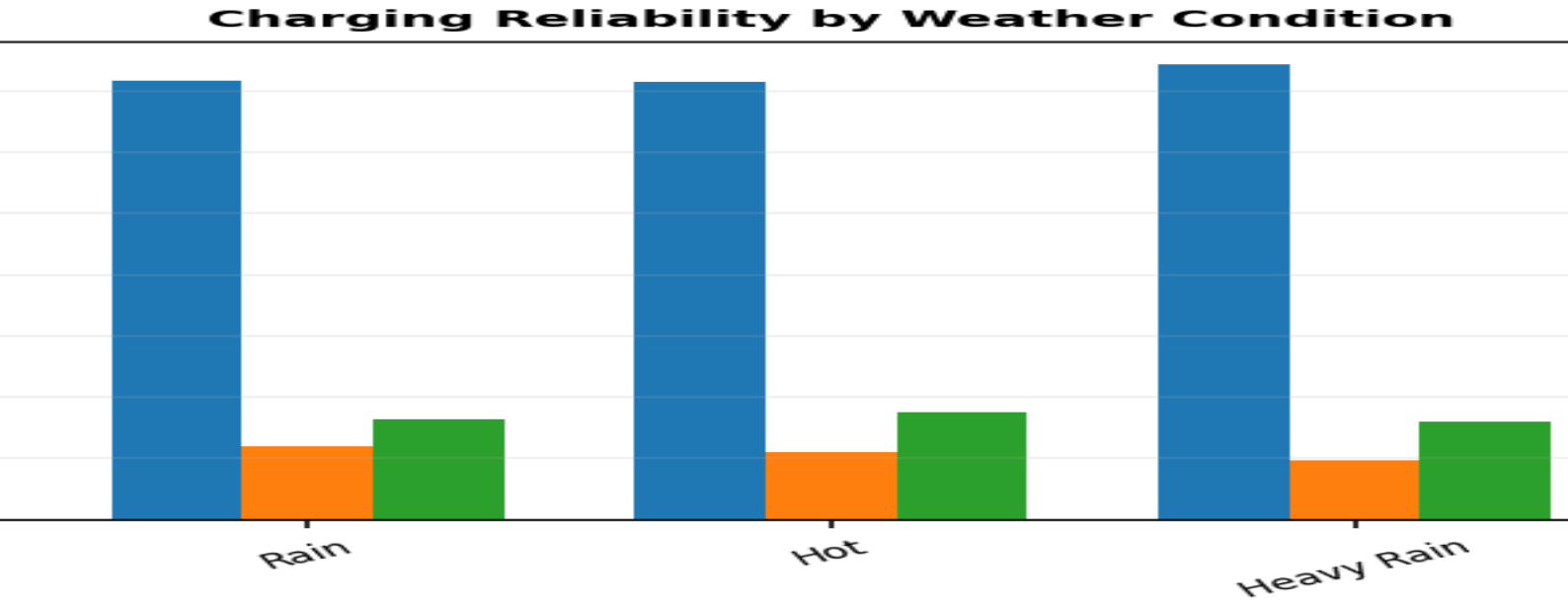
Analysis / Outcome

Highest failure-rate weather group: Clear (13.9%). See the table for interruption and downtime too. Weather groups may differ in sample size and location mix, so this is an association, not proof that weather caused a failure.

Key Insights & Recommended Action

If adverse weather groups show higher risk, inspect drainage, connector protection and preventive maintenance, while controlling for city and charger type.

Visualization



Supporting Results Table

Weather	Total_Sessions	Completion_Rate	Failure_Rate	Interruption_Rate	Avg_Downtime	Avg_Rating	Avg_Energy
Clear	445.00	68.76	13.93	17.30	12.32	3.21	33.11
Rain	444.00	71.62	11.94	16.44	9.52	3.16	33.39
Hot	452.00	71.46	11.06	17.48	11.69	3.18	36.53
Heavy Rain	437.00	74.37	9.61	16.02	9.55	3.20	34.82
Cloudy	436.00	72.02	7.11	20.87	10.94	3.29	35.06

Q8. GenAI-Assisted Vehicle Segment Demand Analysis

Problem Statement

Measure charging demand and business value by vehicle segment.

GenAI Prompt

For each Vehicle_Segment calculate total sessions, total energy, average energy/session, total final revenue, average transaction value, completion rate, failure rate and average rating. Identify demand and value leaders while considering reliability.

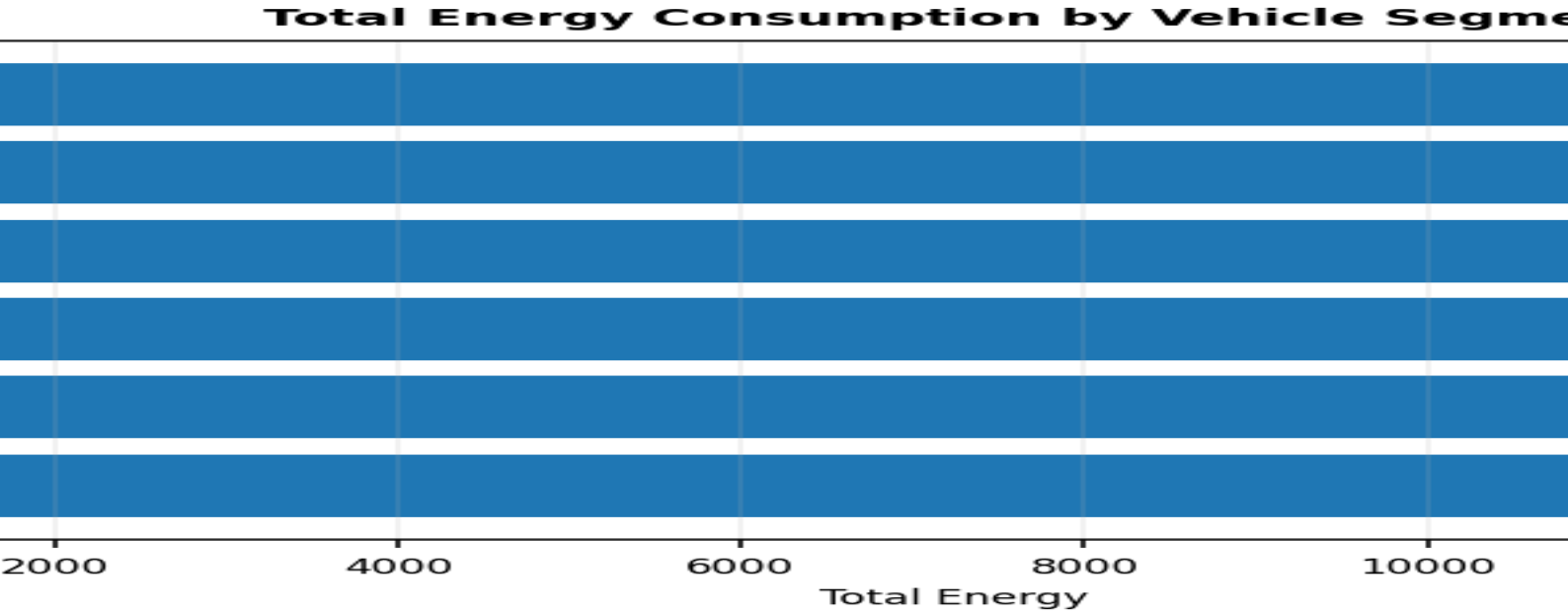
Analysis / Outcome

Highest total energy demand: SUV EV (13,804.43 kWh). Revenue and reliability comparisons appear in the table.

Key Insights & Recommended Action

Use segment demand to plan charger capacity and service windows; total energy reflects volume plus per-session usage and should be read alongside session count.

Visualization



Supporting Results Table

Vehicle_Segment	Total_Sessions	Total_Energy	Avg_Energy	Total_Revenue	Avg_Transaction	Completion_Rate	Failure_Rate	Avg_Rating
SUV EV	401.00	13,804.43	34.95	239,908.83	598.28	71.32	9.73	3.18
Fleet Cab	367.00	12,815.44	36.10	223,409.67	608.75	70.57	10.63	3.27
Delivery Van	370.00	12,398.66	33.78	213,352.66	576.63	71.35	12.43	3.20
Two-Wheeler	354.00	12,306.58	35.57	203,590.72	575.12	74.01	7.91	3.26
Sedan EV	356.00	12,095.86	34.56	214,688.92	603.06	70.22	10.96	3.13
Hatchback EV	366.00	11,828.14	32.58	201,248.20	549.86	72.40	12.84	3.21

Q9. GenAI-Assisted Peak Charging Demand Analysis

Problem Statement

Identify high-demand periods from session start times to inform capacity and staffing.

GenAI Prompt

Extract Session_Hour from Session_Start, assign Night/Morning/Afternoon/Evening using the assignment ranges, and compare sessions, energy, revenue and reliability by time band. Highlight peak period and operational implications.

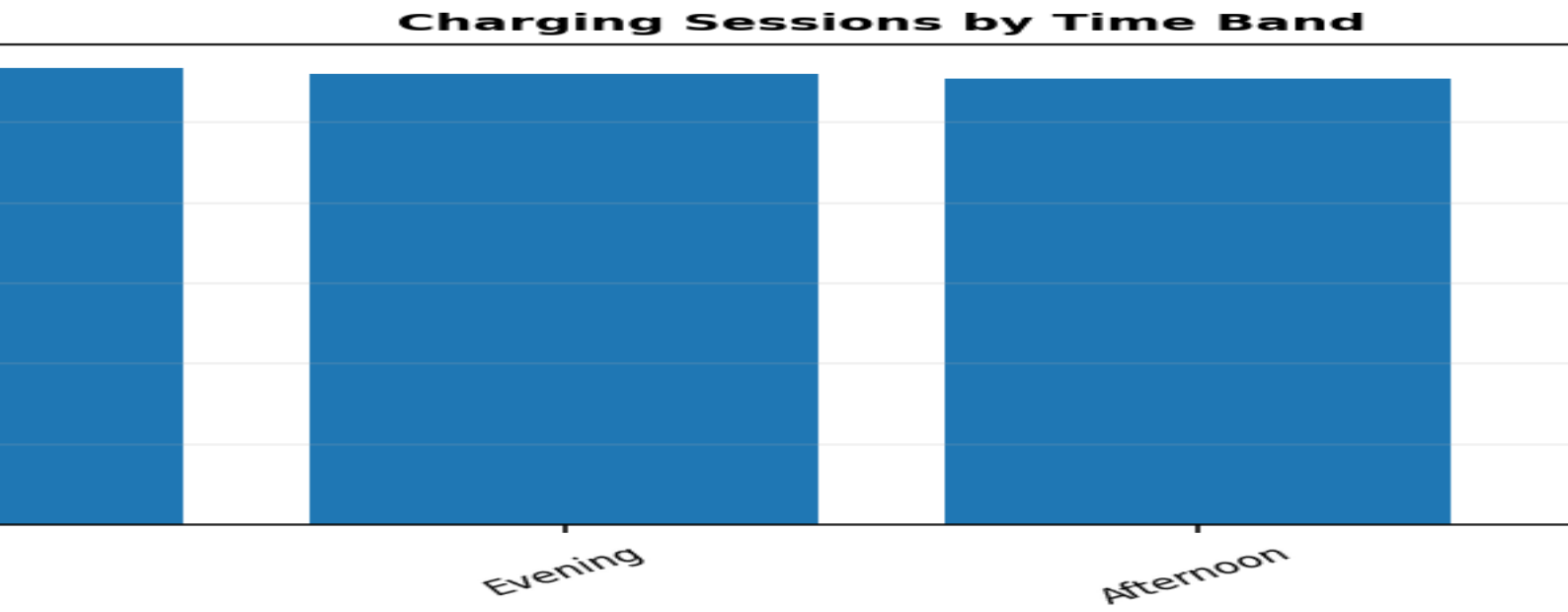
Analysis / Outcome

Highest-volume time band: Night (568 sessions). The chart compares four time bands; unknown start times are retained as Unknown rather than forced into a period.

Key Insights & Recommended Action

Align staffing and preventive checks with the peak band; validate by weekday/weekend and month before changing capacity.

Visualization



Supporting Results Table

Time_Band	Total_Sessions	Total_Energy	Total_Revenue	Completion_Rate	Failure_Rate	Avg_Downtime
Night	568.00	20,304.86	345,533.09	74.65	10.39	10.21
Evening	560.00	19,009.92	323,603.97	70.71	8.93	11.39
Afternoon	555.00	18,092.00	319,877.49	71.89	10.45	10.11
Morning	531.00	17,842.33	307,184.45	69.11	13.37	11.60

Q10. AI-Powered Charging Station Risk Scoring

Problem Statement

Create a 0–100 station risk score using problem-session rate, downtime and customer rating.

GenAI Prompt

Aggregate station sessions, failed/interrupted sessions, problem rate, average downtime and average rating. Min-max normalize adverse indicators. Score = 50% problem-session rate + 25% average downtime risk + 25% inverse rating risk. Map scores to Low (0–33.33), Medium (>33.33–66.66), High (>66.66) and rank the top 10. Treat small sample sizes cautiously.

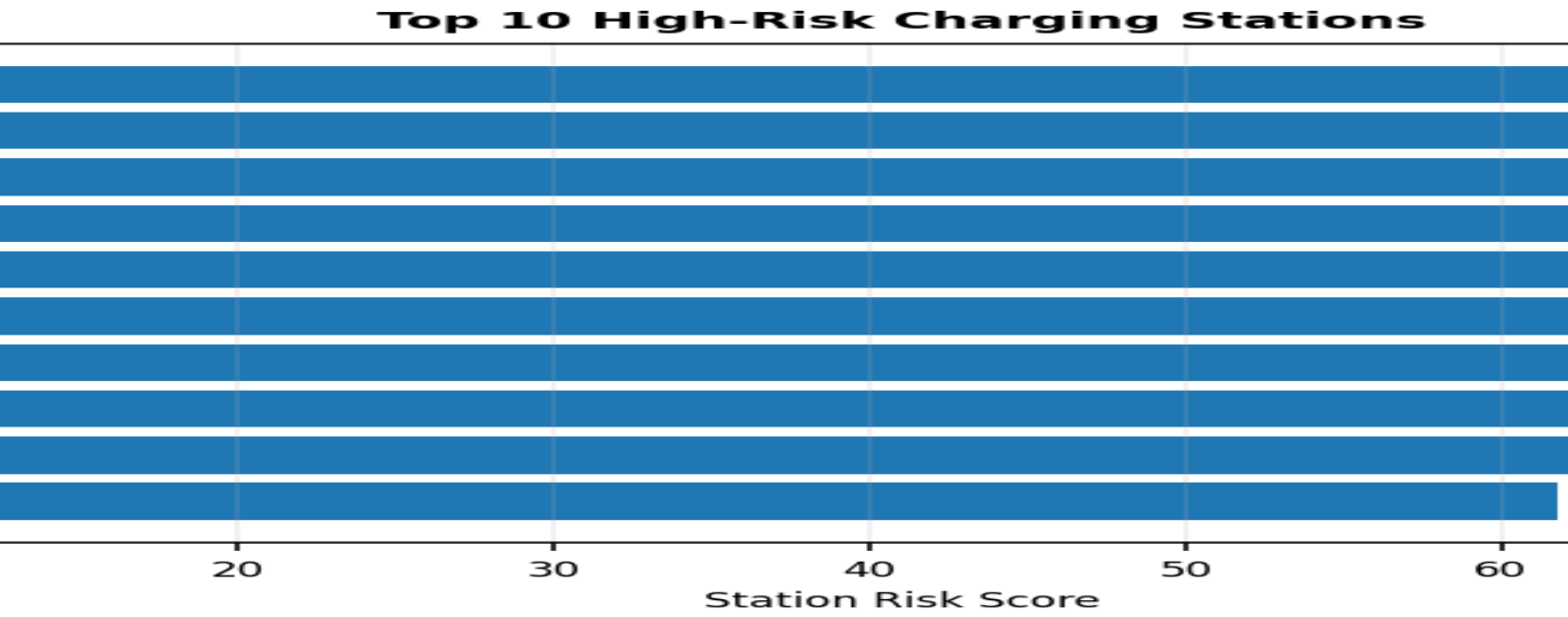
Analysis / Outcome

Top-risk station shown first: EVS0142, score 78.2/100 (High Risk). Weights explicitly prioritize failure/interruption experience (50%) over downtime and rating (25% each).

Key Insights & Recommended Action

Use the list for triage, not as a final verdict. Add a minimum-session threshold and inspect raw incidents before assigning maintenance.

Visualization



Supporting Results Table

Station_ID	Total_Sessions	Problem_Rate	Avg_Downtime	Avg_Rating	Station_Risk_Score	Risk_Category
EVS0142	14.00	64.29	27.09	1.93	78.18	High Risk
EVS0116	7.00	71.43	25.09	2.57	74.18	High Risk
EVS0131	10.00	60.00	24.25	2.50	68.48	High Risk
EVS0163	12.00	50.00	32.19	2.67	68.09	High Risk
EVS0144	10.00	60.00	21.18	2.30	67.97	High Risk
EVS0006	9.00	55.56	22.98	2.56	64.75	Medium Risk
EVS0172	10.00	50.00	21.61	2.30	63.31	Medium Risk
EVS0028	13.00	61.54	13.29	2.25	63.08	Medium Risk
EVS0015	6.00	50.00	19.78	2.20	62.82	Medium Risk
EVS0037	7.00	57.14	21.49	2.83	61.79	Medium Risk

Q11. AI-Powered Charging Operator Performance Benchmarking

Problem Statement

Benchmark operators using reliability, experience, downtime, revenue and energy.

GenAI Prompt

For each Operator calculate sessions, energy, revenue, completion rate, failure rate, downtime and rating. Min-max normalize indicators and calculate score: completion 30%, inverse failure 20%, inverse downtime 15%, rating 20%, revenue 10%, energy 5%. Rank best to worst. Explain that relative min-max scores depend on this dataset.

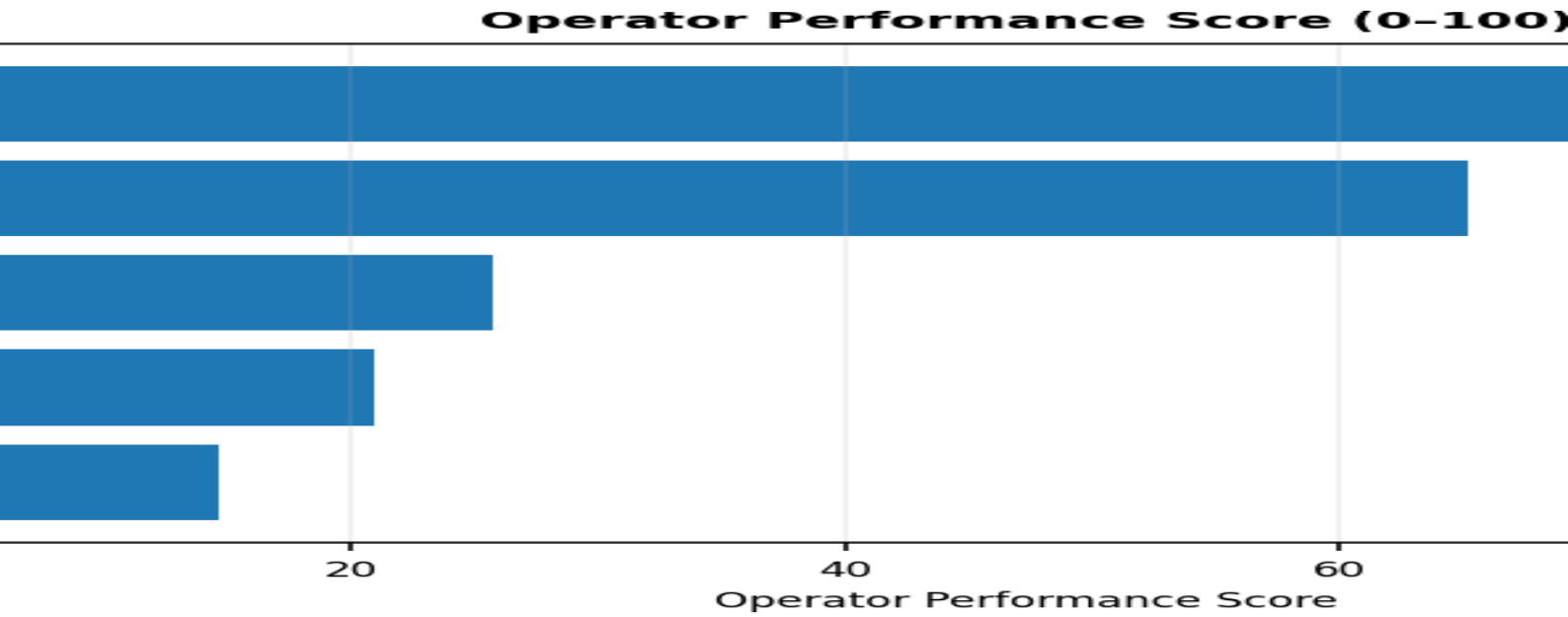
Analysis / Outcome

Highest composite score: VoltWay Mobility (90.2/100). Score weights prioritize reliability and customer experience; revenue and energy contribute only 15% combined.

Key Insights & Recommended Action

Compare score components before decisions: a strong overall score can hide a specific weakness. Recalculate periodically as the network changes.

Visualization



Supporting Results Table

Operator	Total_Sessions	Total_Energy	Total_Revenue	Completion_Rate	Failure_Rate	Avg_Downtime	Avg_Rating	Operator_Perform
VoltWay Mobility	528.00	19,680.79	330,215.40	73.48	9.28	10.18	3.25	90.19
GreenGrid Charge	406.00	14,023.20	236,931.83	73.89	11.33	9.35	3.21	65.16
UrbanPlug Networks	489.00	21,201.67	365,956.26	70.35	11.45	11.00	3.17	25.75
ChargeNest Energy	484.00	12,485.73	222,727.20	70.04	10.54	11.72	3.18	20.96
SparkRoute EV	307.00	7,857.72	140,368.31	70.03	11.73	12.12	3.23	14.64

Q12. GenAI-Assisted Charger Age & Maintenance Pattern Analysis

Problem Statement

Check whether installation year is associated with reliability and downtime.

GenAI Prompt

Group by Installation_Year and calculate sessions, completion, failure, problem rate, average downtime and rating. Compare results without assuming older installations perform worse; consider sample sizes and confounders.

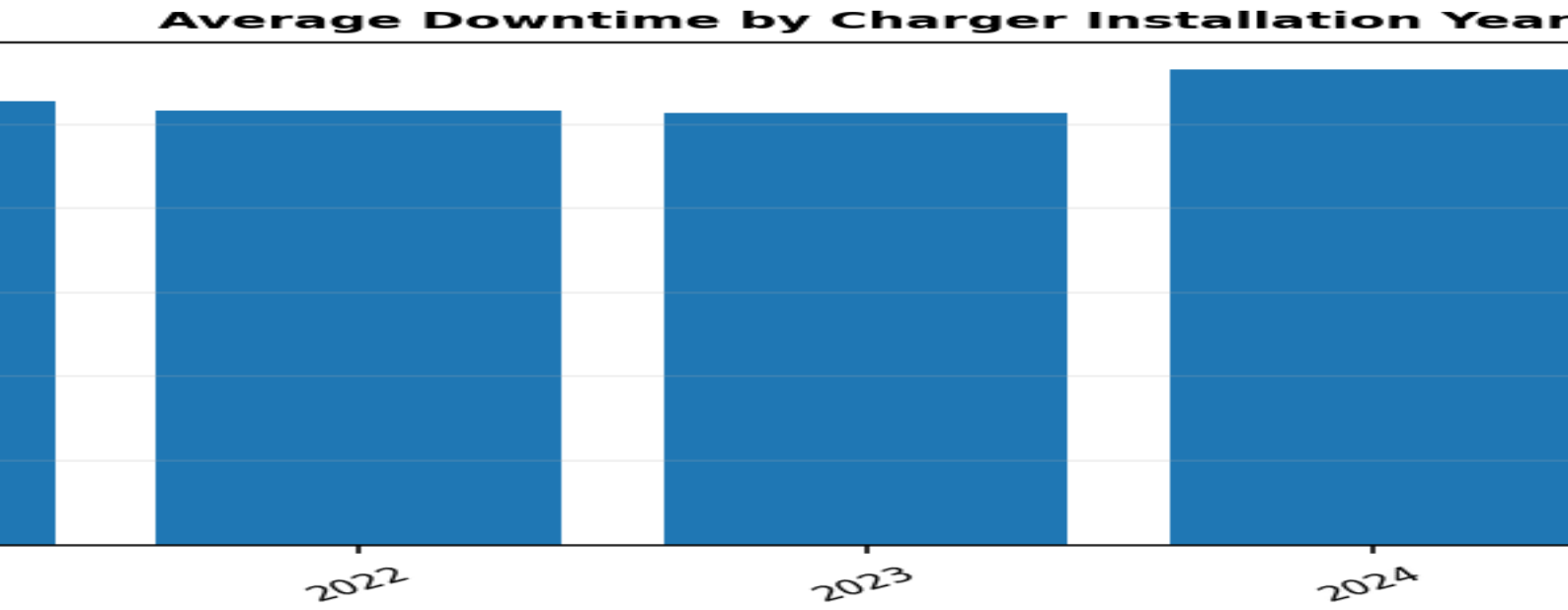
Analysis / Outcome

Highest average downtime installation year: 2025 (11.38 minutes). The line/bar comparison shows observed patterns; it does not establish age as the cause.

Key Insights & Recommended Action

Use year-level results to target inspections, but validate against charger model, usage intensity and recent maintenance history.

Visualization



Supporting Results Table

Installation_Year	Total_Sessions	Completion_Rate	Failure_Rate	Problem_Rate	Avg_Downtime	Avg_Rating
2,021.00	328.00	71.95	10.37	28.05	10.56	3.27
2,022.00	384.00	69.79	12.24	30.21	10.32	3.15
2,023.00	482.00	72.61	11.00	27.39	10.27	3.26
2,024.00	496.00	70.36	11.90	29.64	11.30	3.20
2,025.00	524.00	73.09	8.59	26.91	11.38	3.17

Q13. GenAI-Assisted Customer Access Channel Experience Analysis

Problem Statement

Compare channels on volume, revenue, completion/failure, problem rate, ratings and downtime.

GenAI Prompt

Group by Source_Channel and calculate sessions, revenue, average transaction value, completion rate, failure rate, problem rate, average rating and downtime. Use a scatter plot of problem rate vs rating to compare operational reliability and customer experience.

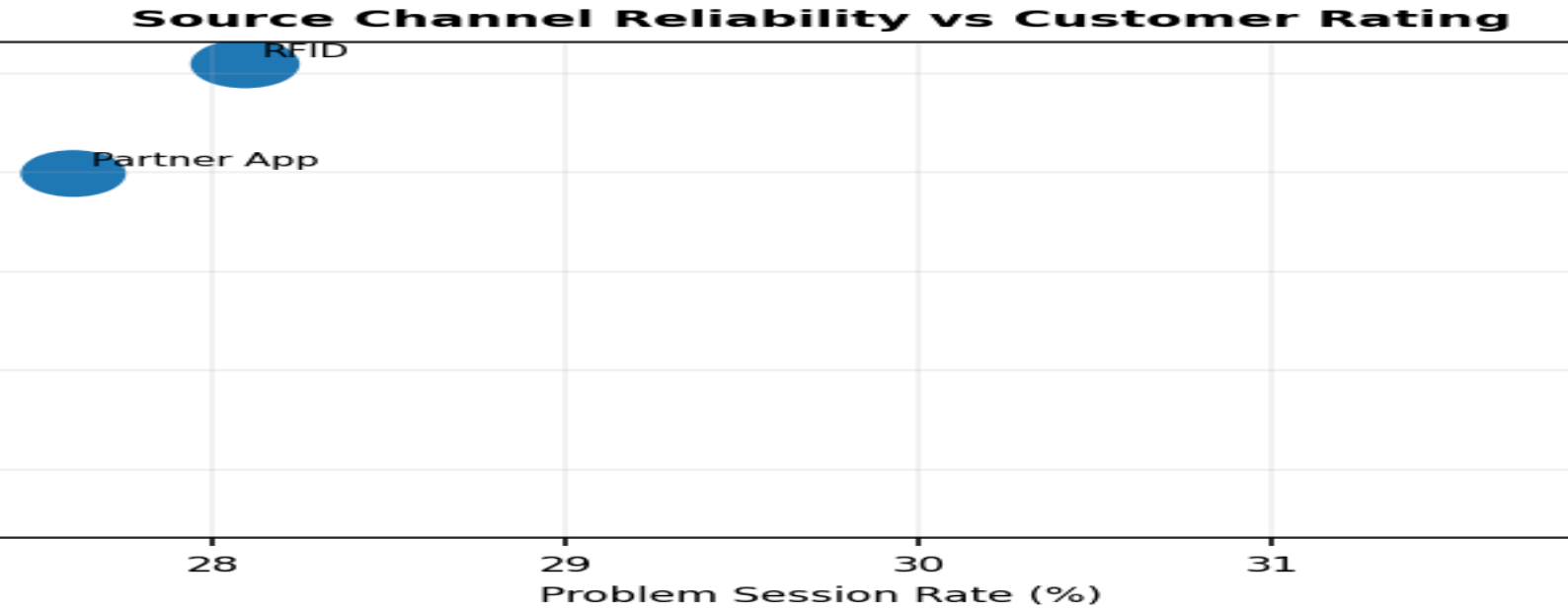
Analysis / Outcome

Best combined completion/rating channel: Mobile App. Highest problem-session rate: Walk-in QR (33.0%). See chart and table for context.

Key Insights & Recommended Action

Investigate channel-specific friction such as onboarding, app/network reliability or payment handoffs. Avoid interpreting association as causation.

Visualization



Supporting Results Table

Source_Channel	Total_Sessions	Total_Revenue	Avg_Transaction	Completion_Rate	Failure_Rate	Problem_Rate	Avg_Rating	Avg_Downtime
Mobile App	453.00	269,910.78	595.83	73.51	8.83	26.49	3.25	10.89
Fleet API	466.00	262,501.91	563.31	73.18	11.59	26.82	3.17	10.53
Partner App	413.00	229,374.98	555.39	72.40	9.20	27.60	3.25	9.63
RFID	445.00	262,574.07	590.05	71.91	11.24	28.09	3.30	9.69
Walk-in QR	437.00	271,837.26	622.05	67.05	12.81	32.95	3.08	13.31

Q14. GenAI-Assisted Technician Performance & Workload Analysis

Problem Statement

Compare workload with reliability, downtime, ratings and a transparent effectiveness score.

GenAI Prompt

For each Technician calculate sessions handled, completion rate, problem-session rate, average downtime and average rating. Score = completion 35%, inverse problem rate 25%, inverse normalized downtime 20%, normalized rating (/5) 20%. Compare workload against completion rate and do not infer workload causes poor performance.

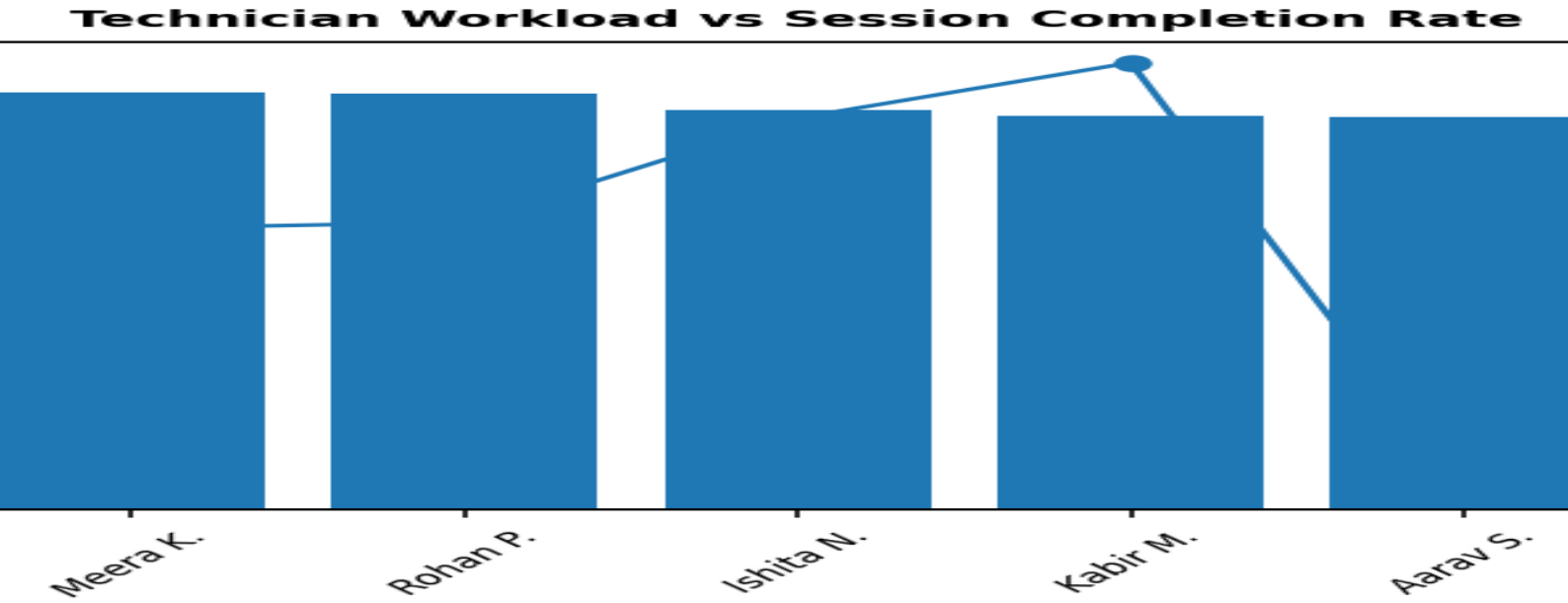
Analysis / Outcome

Highest effectiveness score: Ishita N. (62.9/100). Lowest score: Aarav S. (52.5/100). Score uses available ratings and downtime; missing values are median-filled only for the composite score.

Key Insights & Recommended Action

Use results to guide coaching and workload balancing, not to penalize technicians without reviewing case complexity and assignment context.

Visualization



Supporting Results Table

Technician	Total_Sessions	Completion_Rate	Problem_Rate	Avg_Downtime	Avg_Rating	Technician_Effectivenes
Ishita N.	354.00	73.73	26.27	9.31	3.21	62.93
Kabir M.	349.00	74.79	25.21	9.68	3.19	62.92
Neha R.	395.00	71.39	28.61	10.30	3.24	60.16
Meera K.	370.00	71.62	28.38	11.01	3.20	59.02
Rohan P.	368.00	71.74	28.26	11.30	3.29	59.02
Unknown	30.00	70.00	30.00	12.90	3.00	54.40
Aarav S.	348.00	66.67	33.33	13.16	3.13	52.52

Q15. AI-Powered Customer Experience Index by City

Problem Statement

Combine completion, problem-session rate, downtime and ratings into a city-level experience score.

GenAI Prompt

Normalize city indicators to 0–1. Score = completion 35%, rating 30%, inverse problem rate 20%, inverse downtime 15%. Rank cities and identify improvement opportunities. Explain that min-max scoring is relative to the current cities and weights are assumptions.

Analysis / Outcome

Highest score city: bengaluru (100.0/100). Lowest score city: delhi ncr (0.0/100).

Key Insights & Recommended Action

Use score to focus improvement projects, and review underlying rates because a composite score can mask individual problems.

Visualization



Supporting Results Table

City	Total_Sessions	Completion_Rate	Problem_Rate	Avg_Downtime	Avg_Rating	Customer_Experience_S
bengaluru	9.00	88.89	11.11	1.99	4.11	100.00
BENGALURU	4.00	75.00	25.00	7.35	3.50	68.66
Jaipur	262.00	77.48	22.52	8.94	3.27	66.63
Unknown	22.00	77.27	22.73	18.75	3.27	62.04
Mumbai	402.00	73.38	26.62	10.24	3.29	61.39
Delhi	177.00	71.75	28.25	11.28	3.24	57.83
Bangalore	230.00	72.61	27.39	10.46	3.16	57.69
Pune	326.00	70.55	29.45	10.04	3.24	57.00
Chennai	249.00	70.28	29.72	10.53	3.18	55.29
Hyderabad	297.00	69.02	30.98	12.30	3.09	51.08
Ahmedabad	216.00	67.59	32.41	12.02	3.14	50.39
DELHI NCR	7.00	57.14	42.86	15.53	3.00	33.30

Q16. GenAI-Assisted Charging Demand Forecasting

Problem Statement

Summarize monthly demand and forecast the next three months using complete historical months.

GenAI Prompt

Aggregate complete historical months for sessions, energy, revenue and completion rate. Fit a simple linear trend to monthly session counts and forecast the next 3 months. Distinguish forecast from actuals and state that a short-history linear baseline is directional, not a production forecast.

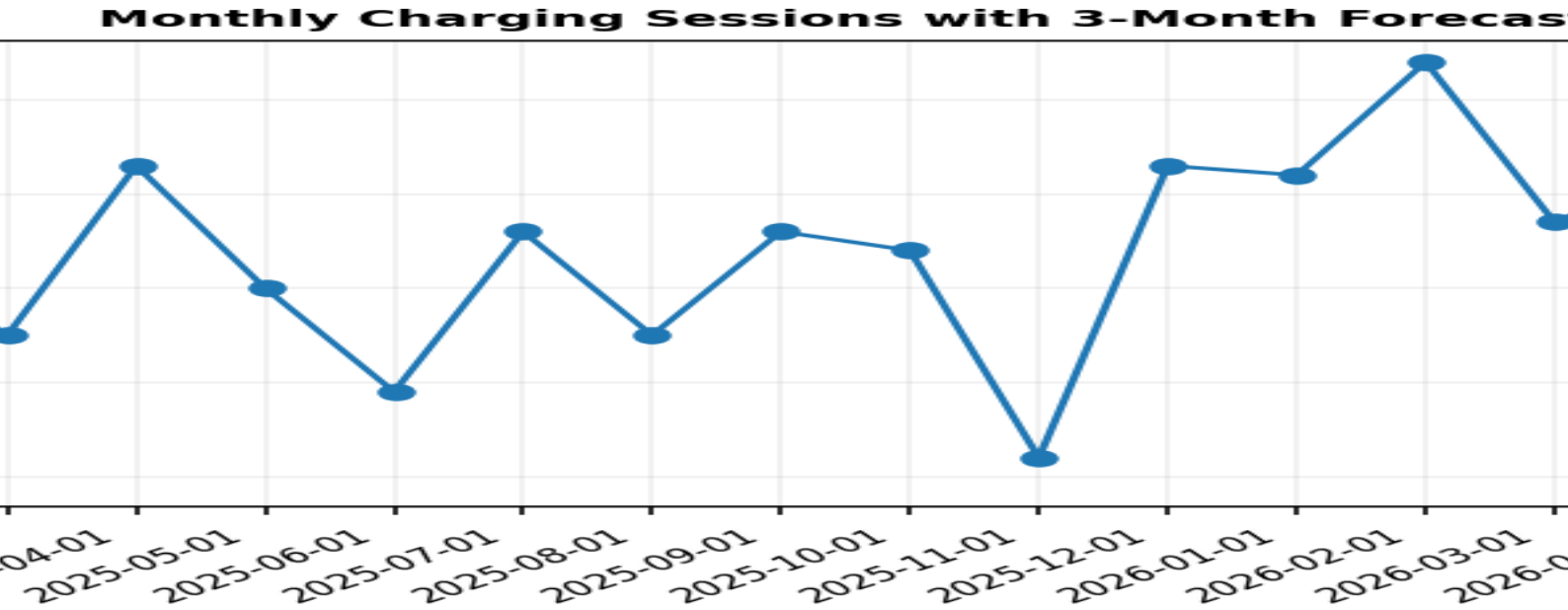
Analysis / Outcome

Historical monthly demand appears increasing. Forecast sessions: Jun 2026: 133, Jul 2026: 134, Aug 2026: 135. Forecast uses a linear trend over complete months only; no seasonality, holidays, charger expansion or external factors are modeled.

Key Insights & Recommended Action

Use the forecast as an initial capacity-planning signal, with safety buffer and frequent refresh. Do not treat it as a guaranteed demand target.

Visualization



Supporting Results Table

Month	Total_Sessions	Total_Energy	Total_Revenue	Completion_Rate
2025-10-01 00:00:00	126.00	4,314.27	73,817.60	72.22
2025-11-01 00:00:00	124.00	3,642.02	61,814.65	70.97
2025-12-01 00:00:00	102.00	3,850.99	61,096.71	67.65
2026-01-01 00:00:00	133.00	4,588.31	81,800.43	72.18
2026-02-01 00:00:00	132.00	4,652.35	83,625.88	72.73
2026-03-01 00:00:00	144.00	4,701.89	77,299.40	69.44
2026-04-01 00:00:00	127.00	4,437.33	77,418.14	70.87
2026-05-01 00:00:00	133.00	4,413.93	78,672.74	75.94
2026-06-01 00:00:00	133.00	—	—	—
2026-07-01 00:00:00	134.00	—	—	—
2026-08-01 00:00:00	135.00	—	—	—

Q17. GenAI-Assisted EV Connector Demand Mix Analysis

Problem Statement

Quantify connector usage, share, energy and revenue.

GenAI Prompt

For each Connector_Type calculate session count, share of total sessions, total energy, average energy/session and total final revenue. Use session share for the pie chart and assess concentration versus diversification.

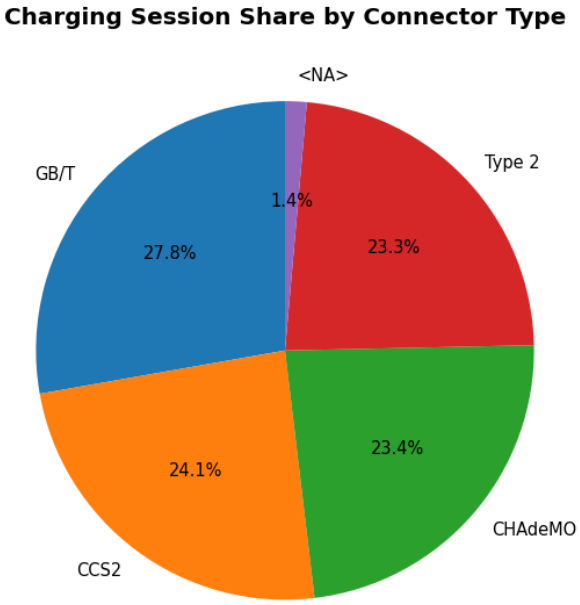
Analysis / Outcome

Most frequently used connector: GB/T (27.8% of sessions). The connector table gives energy and revenue alongside share.

Key Insights & Recommended Action

Prioritize compatibility and availability for high-share connectors while maintaining support for lower-share standards where customer or fleet requirements justify it.

Visualization



Supporting Results Table

Connector_Type	Total_Sessions	Session_Share_%	Total_Energy	Avg_Energy	Total_Revenue
GB/T	615.00	27.78	20,395.75	33.66	348,622.98
CCS2	534.00	24.12	17,053.39	32.36	293,032.73
CHAdeMO	519.00	23.44	16,882.40	33.36	300,506.81
Type 2	515.00	23.26	19,720.65	38.97	333,809.90
Unknown	31.00	1.40	1,196.92	38.61	20,226.58

Q18. AI-Generated Executive Dashboard & Management Report

Problem Statement

Provide executive KPIs and the six required visuals in a single dashboard.

GenAI Prompt

Design an executive EV operations dashboard from the cleaned and enriched dataset. Include KPI cards for total sessions, total energy, total revenue, completion rate and average rating; visuals for monthly sessions, revenue by city, sessions by charger type, status distribution, top 10 station risk and connector share. Add a concise management summary. Do not use values not present in the cleaned dataset.

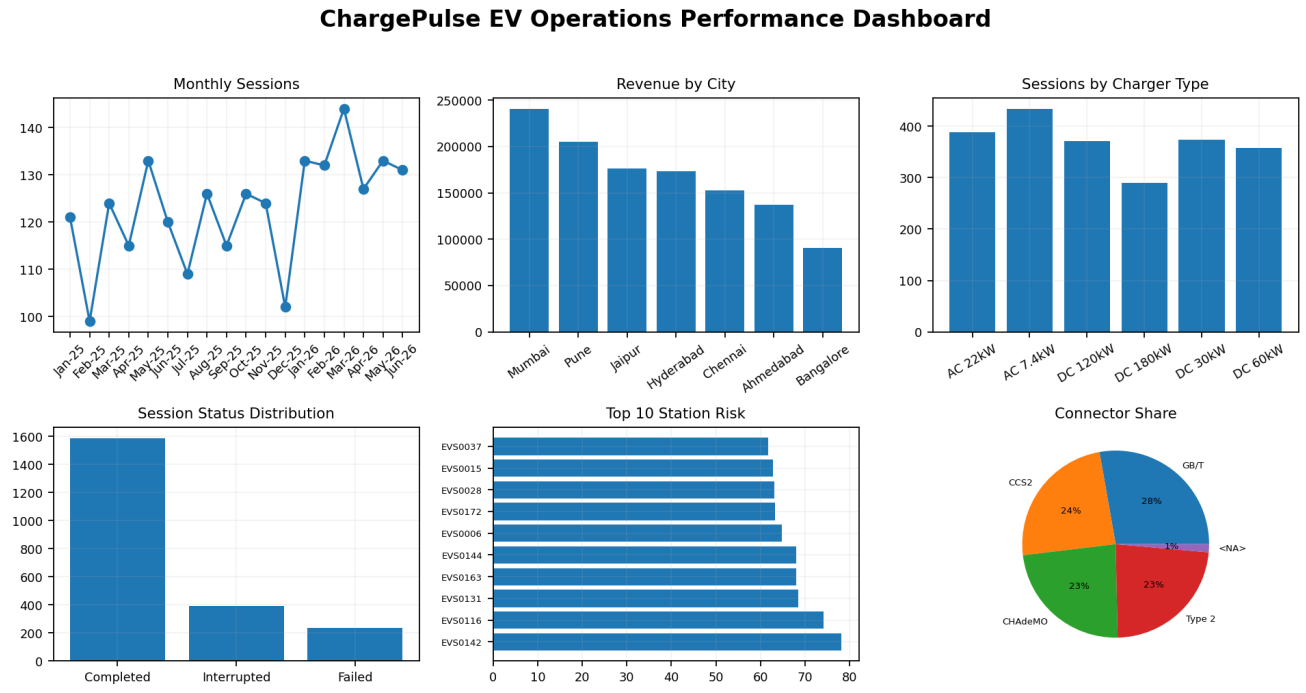
Analysis / Outcome

KPI cards: 2,214 sessions | 75,249.11 kWh delivered | ■1,296,199.00 final revenue | 71.6% completion | 3.21/5 average rating. Dashboard visuals follow. Executive summary: focus first on high-risk stations and city/channel reliability; align capacity with observed time-band and monthly demand; preserve connector availability according to demand mix; investigate issue categories and repeat failures. These are data-led priorities, not causal conclusions.

Key Insights & Recommended Action

The dashboard summarizes demand, revenue, reliability, customer experience and station risk. All values are computed from the cleaned analytical table described in Q1.

Visualization



GENAI PROMPT & OUTPUT EVIDENCE

ChargePulse EV Practical Analytics | Aditi Yadav

Evidence snapshots reconstructed from the documented prompts and analysis in this report.

PROMPT SNAPSHOT — Q1: DATA QUALITY AUDIT

Act as a senior data-quality analyst. Audit the EV_Charging_Raw_Data sheet. Report missing values by column, exact duplicate rows, duplicate Session_ID values, inconsistent City/Payment_Method/Session_Status labels, invalid Energy_kWh (<=0 or >180), Duration_Min (<=0 or >720), Customer_Rating (outside 1–5), and mixed Session_Date_Raw formats. Do not invent values. Recommend transparent cleaning actions and preserve raw columns.

OUTPUT SNAPSHOT — Q1: CLEANING SUMMARY

Raw records: 2,232
Exact duplicate rows removed: 18
Repeated Session_ID rows removed: 0
Final analytical records: 2,214

Invalid numeric values were marked missing under the stated validation rules. Dates were parsed day-first with Session_Start as fallback when needed; obvious category spelling/case variants were standardized.

Interpretation: These are readable evidence snapshots of the prompts and outputs documented in the assignment, not screenshots captured from the ChatGPT application interface.

GENAI EVIDENCE — FEATURE ENGINEERING & FORECAST

PROMPT SNAPSHOT — FEATURE ENGINEERING

Using Energy_kWh, Duration_Min, Final_Amount, Gross_Amount and Session_Start, define useful analytical features while preserving the original raw fields. State formulas, handle invalid or missing inputs transparently, and explain how the features support EV charging operations analysis.

OUTPUT SNAPSHOT — FEATURE ENGINEERING

Charging speed (kWh/hour) = Energy_kWh ÷ (Duration_Min / 60), where valid duration is positive.
Discount amount = Gross_Amount – Final_Amount, where both amounts are available.
Derived values should remain missing if the required inputs are missing or invalid; raw columns are retained for traceability.

PROMPT SNAPSHOT — Q16: DEMAND FORECAST

Aggregate complete historical months for sessions, energy, revenue and completion rate. Fit a simple linear trend to monthly session counts and forecast the next 3 months. Distinguish forecast from actuals and state that a short-history linear baseline is directional, not a production forecast.

Documented forecast outcome: Jun 2026: 133 sessions; Jul 2026: 134; Aug 2026: 135. This is a directional linear-trend baseline; seasonality, holidays, charger expansion and external factors are not modeled.

GENAI EVIDENCE — RISK SCORING & DASHBOARD

PROMPT SNAPSHOT — RISK & OPERATIONS PRIORITIES

Create a transparent station-risk score using the cleaned dataset. Explain the selected indicators, normalize inputs where needed, state the weights and limitations, and avoid treating the score as a causal explanation. Summarize practical actions for high-risk stations.

OUTPUT SNAPSHOT — EXECUTIVE DASHBOARD KPIs

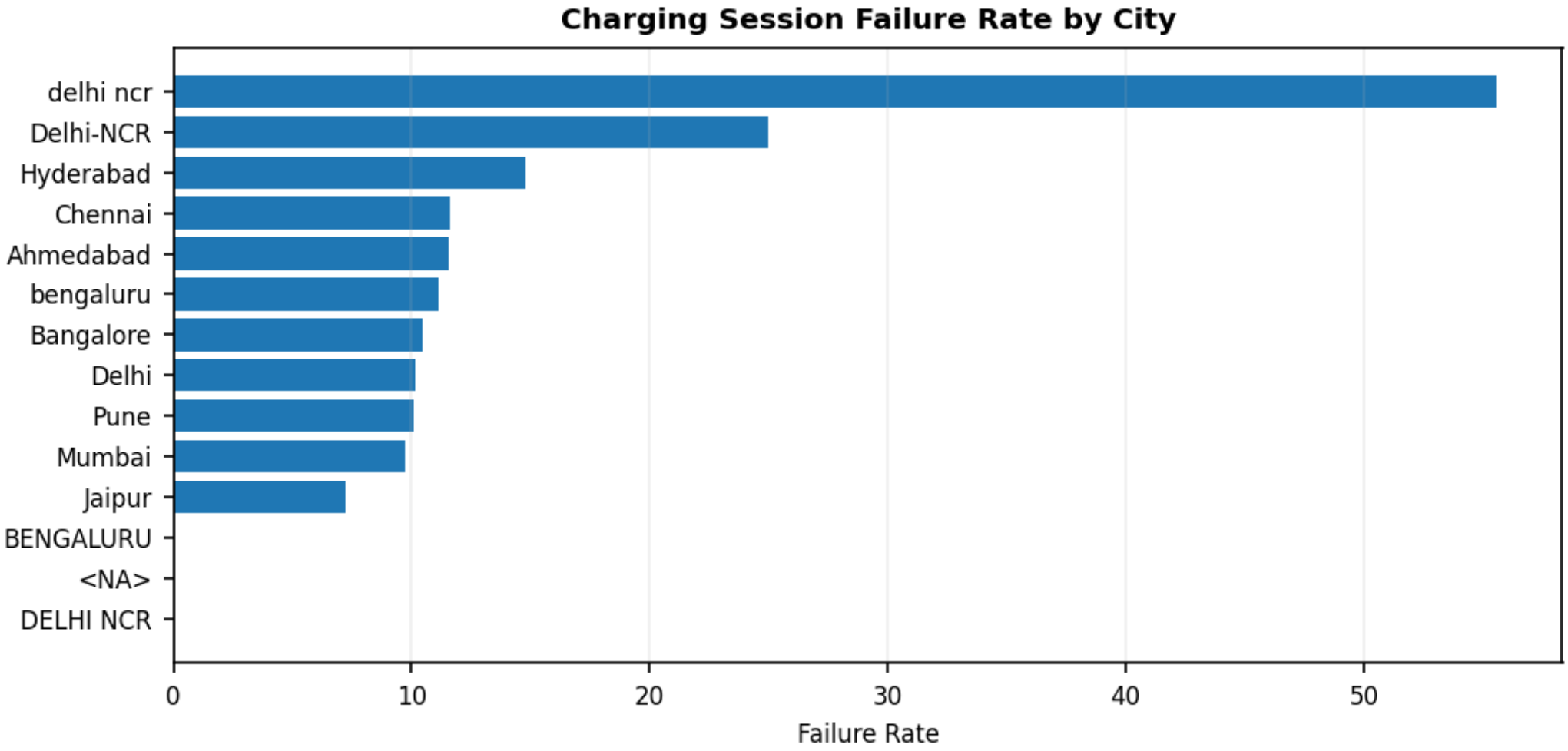
Total sessions: 2,214
Total energy: 75,249.11 kWh
Total final revenue: ■1,296,199.00
Completion rate: 71.6%
Average customer rating: 3.21 / 5

PROMPT SNAPSHOT — Q18: MANAGEMENT DASHBOARD

Design an executive EV operations dashboard from the cleaned and enriched dataset. Include KPI cards for total sessions, total energy, total revenue, completion rate and average rating; visuals for monthly sessions, revenue by city, sessions by charger type, status distribution, top 10 station risk and connector share. Add a concise management summary. Do not use values not present in the cleaned dataset.

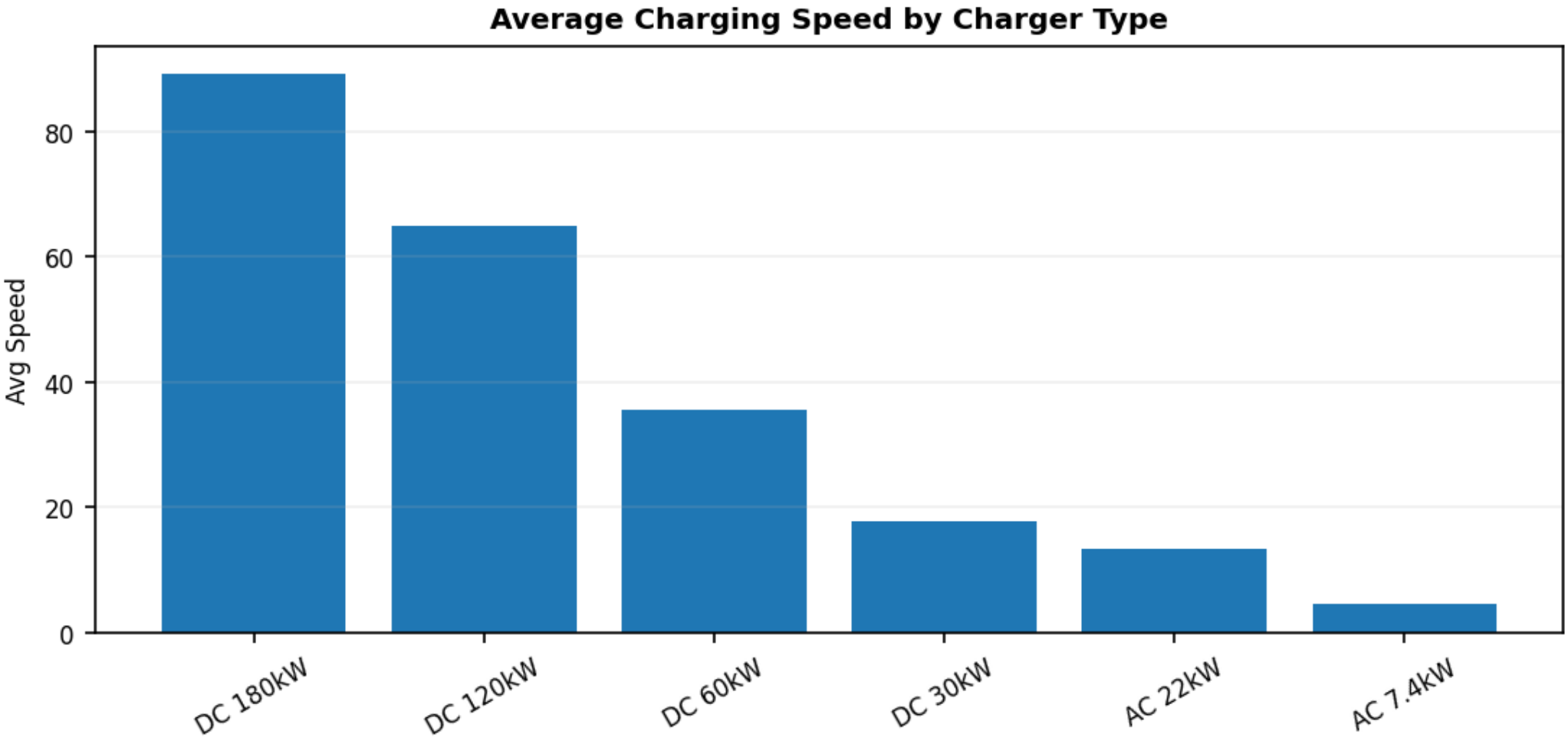
Management interpretation: Prioritize high-risk stations and reliability investigations; align capacity with observed demand; maintain connector availability according to usage mix. These are data-led priorities, not causal conclusions.

Q3. Charging Session Failure Rate by City



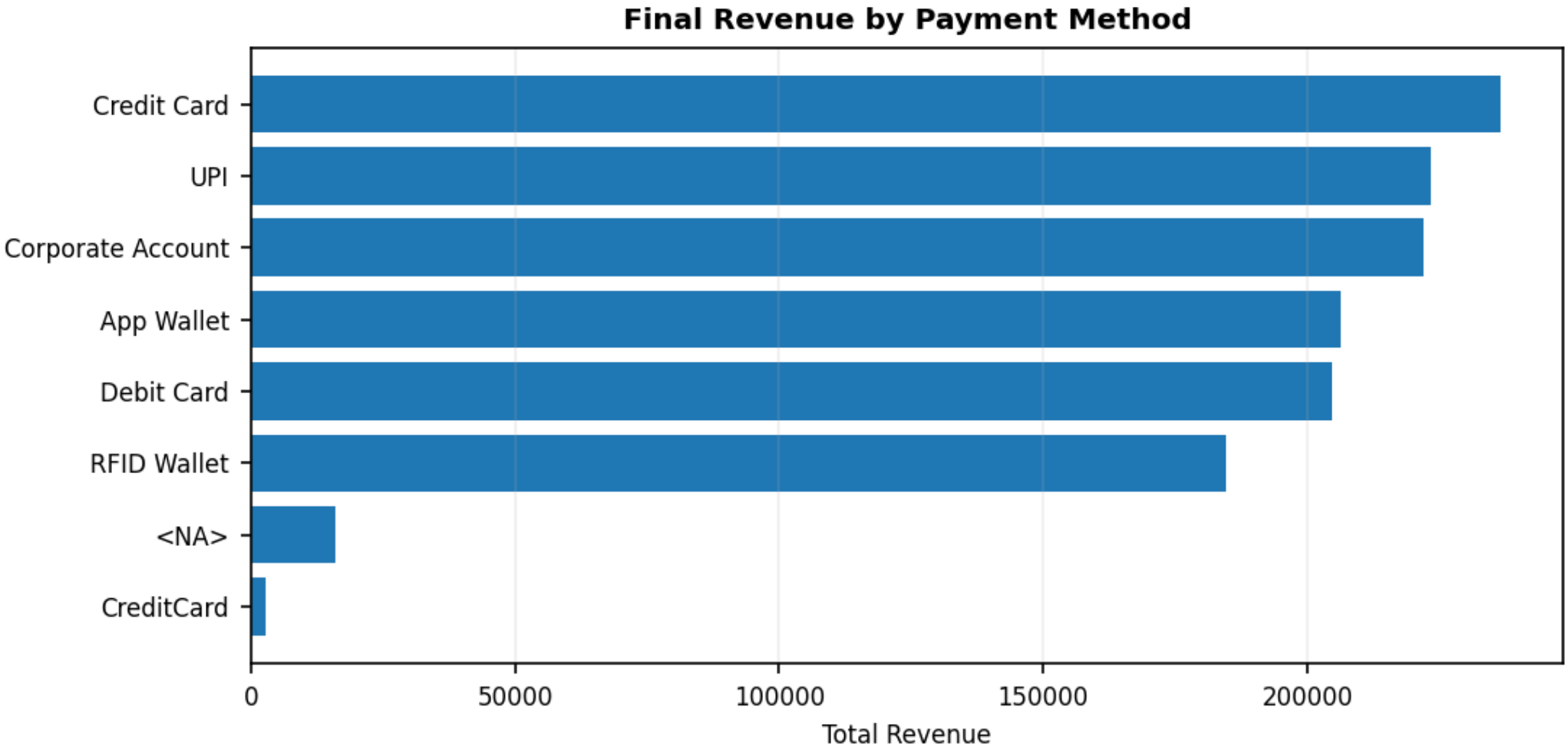
Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q4. Average Charging Speed by Charger Type



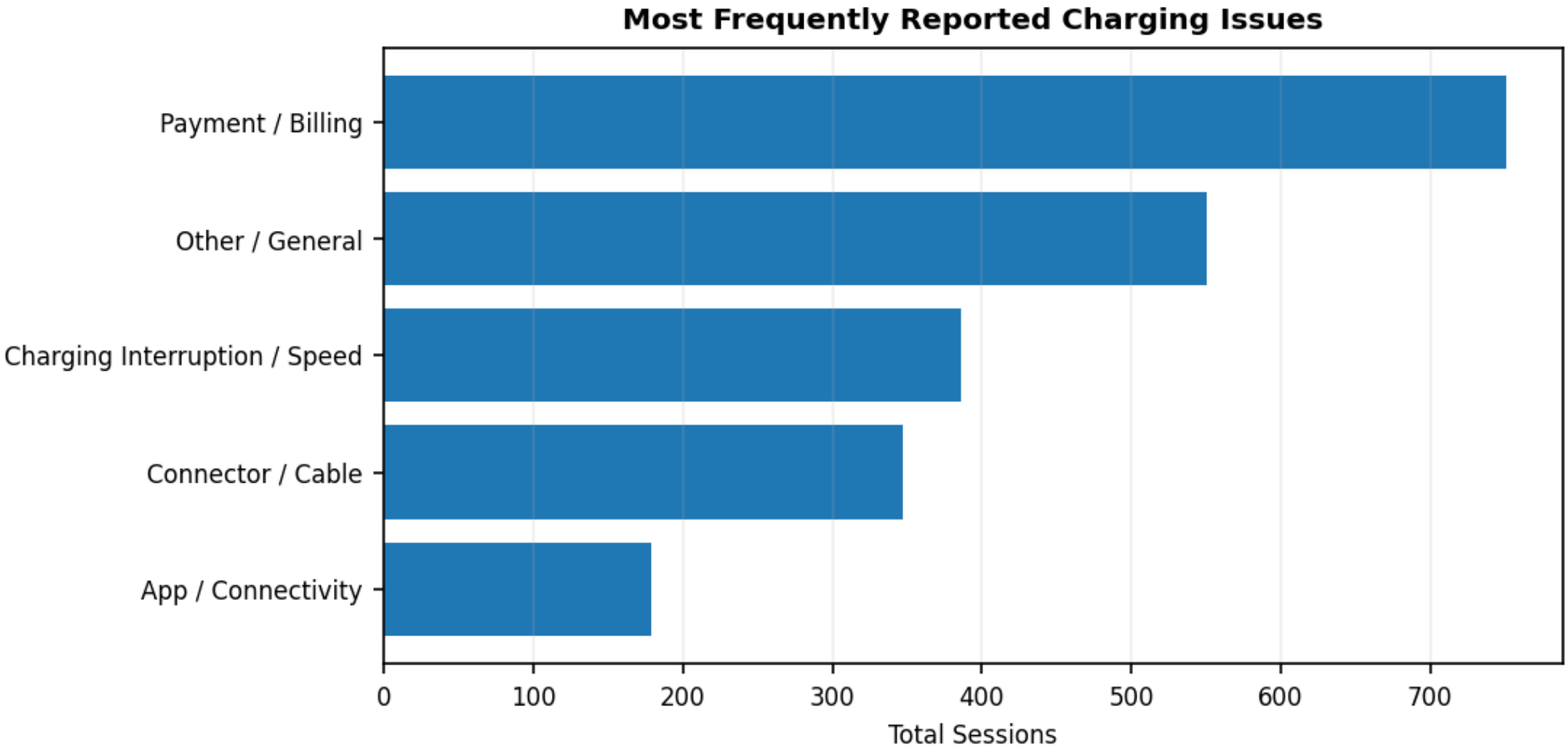
Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q5. Final Revenue by Payment Method



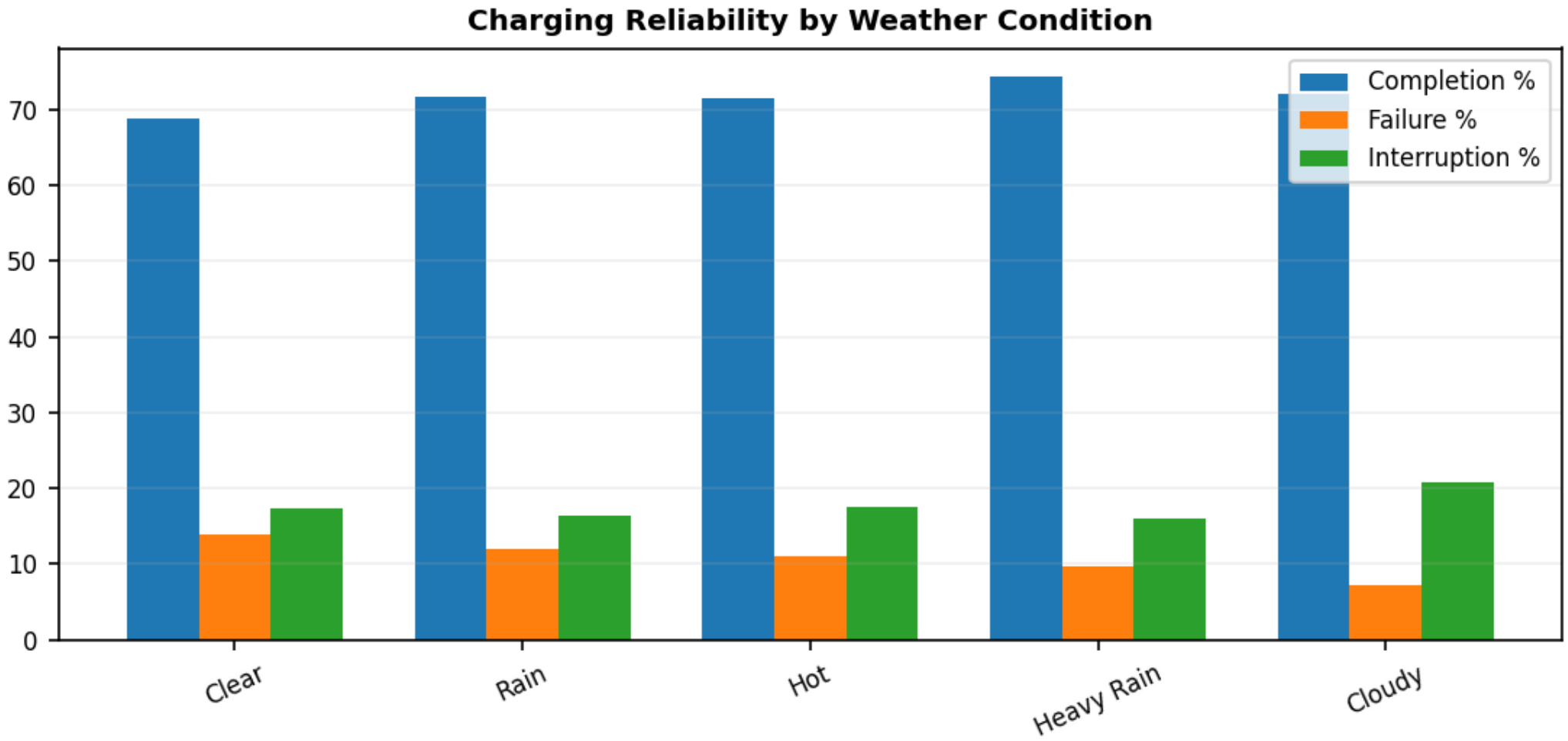
Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q6. Most Frequently Requested Charging Features



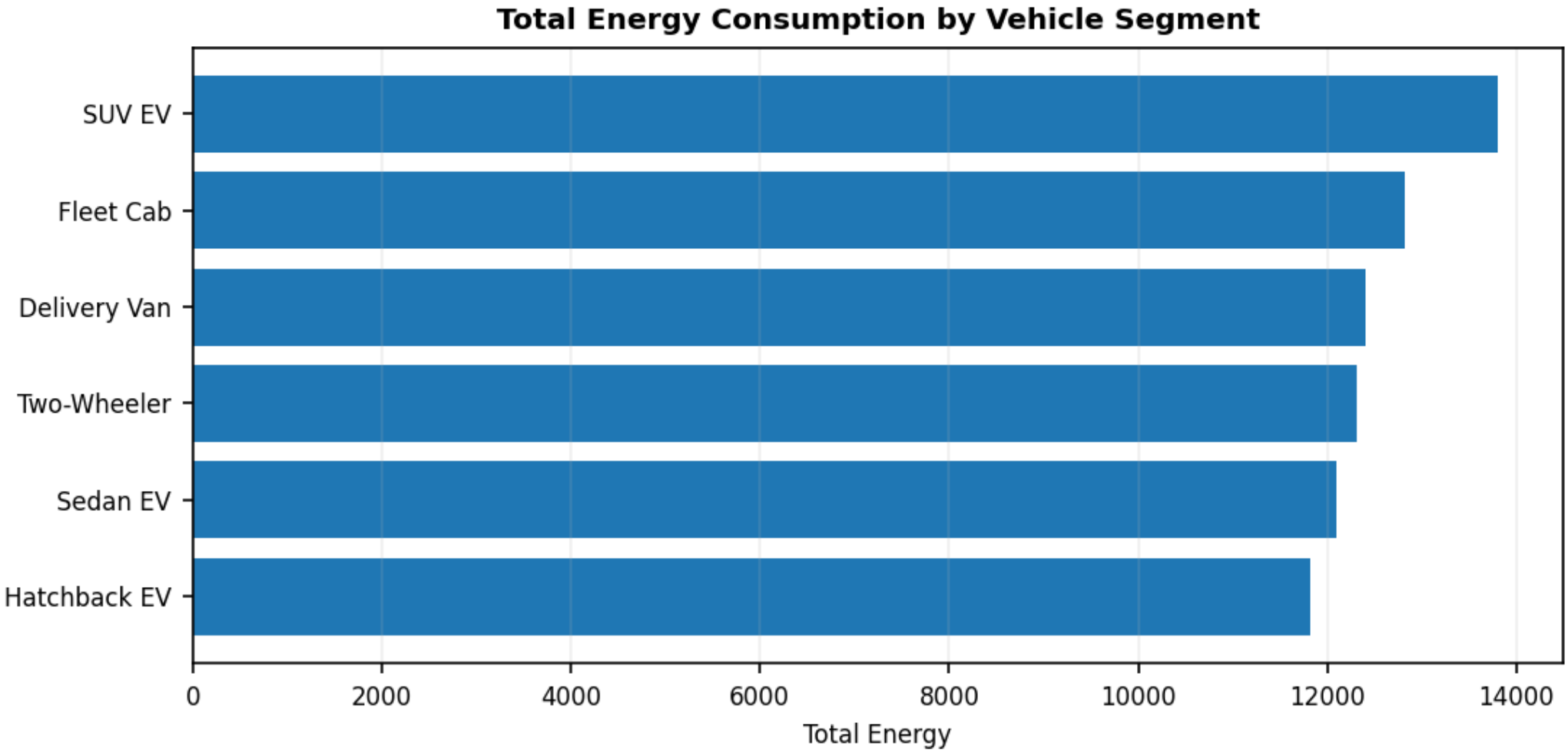
Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q7. Total Energy Consumption by Vehicle Segment



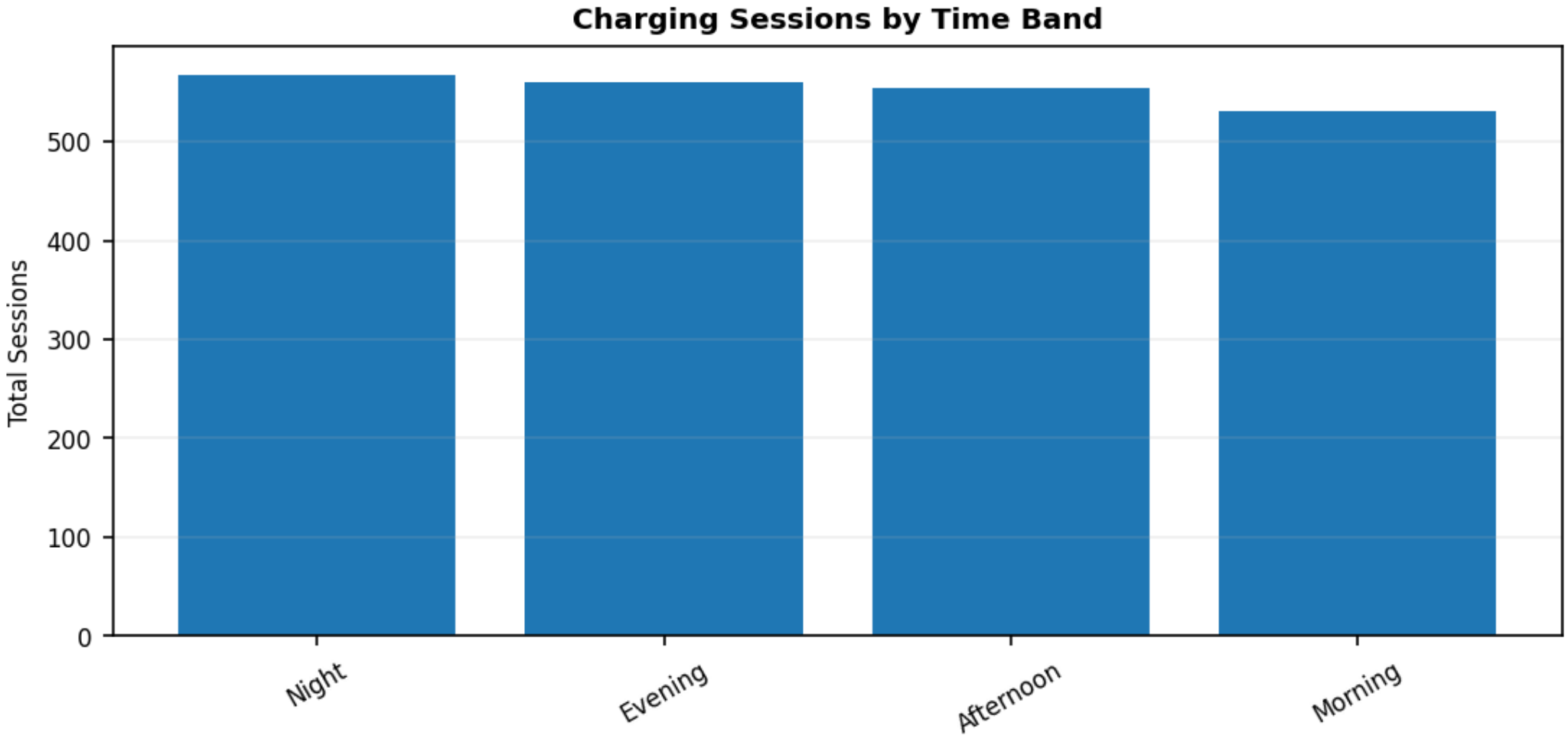
Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q8. Charging Reliability by Weather Condition



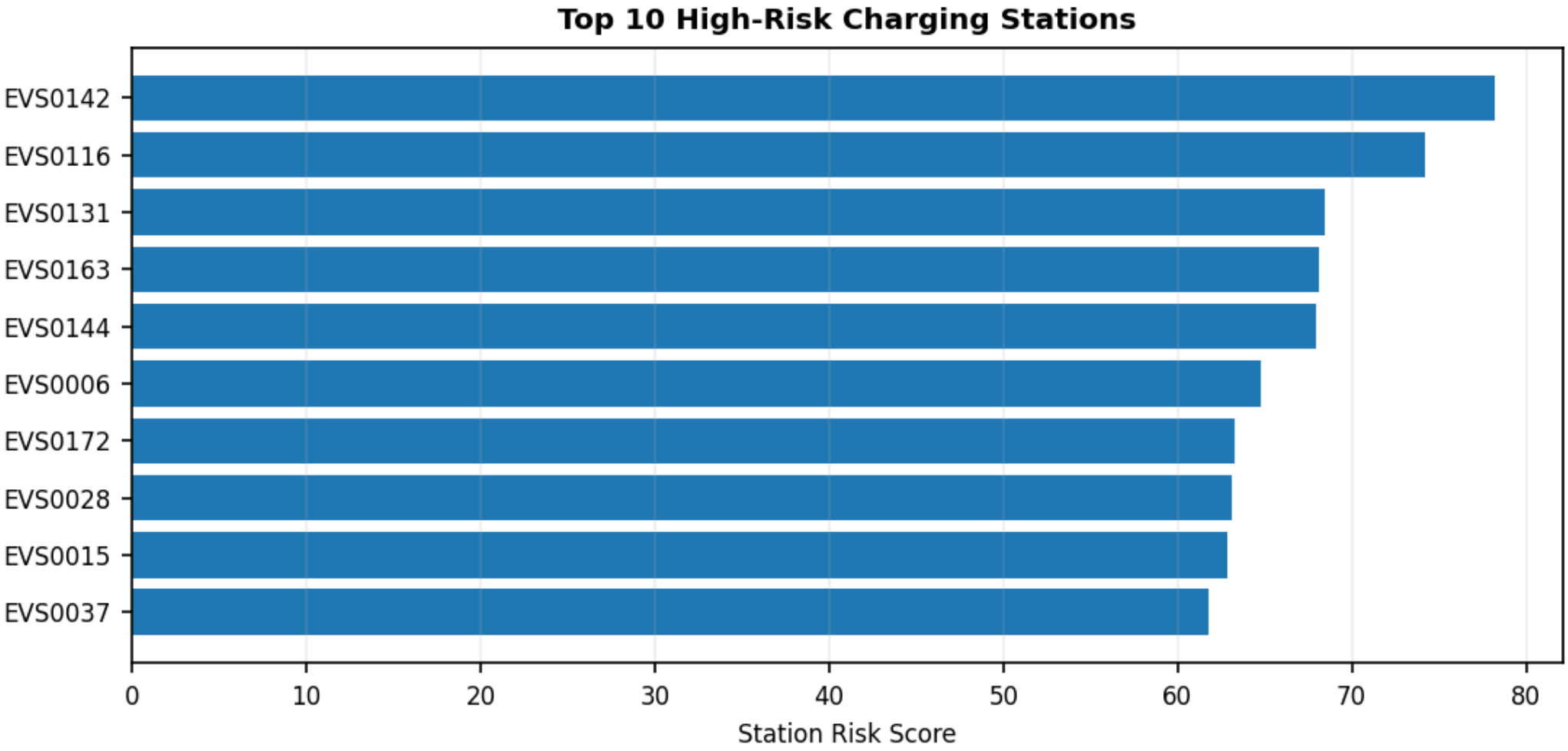
Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q9. Charging Sessions by Time of Day



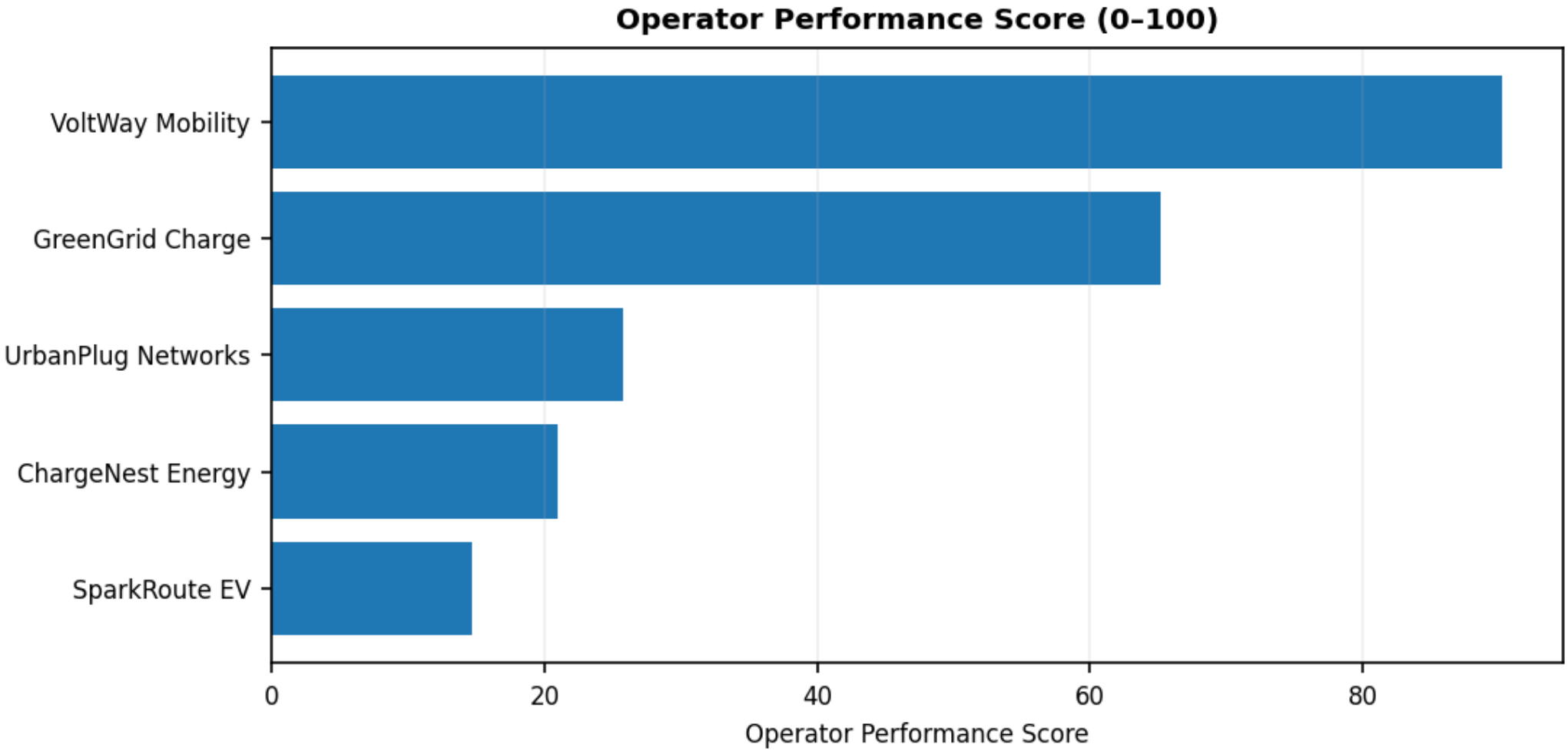
Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q10. Top 10 High-Risk Charging Stations



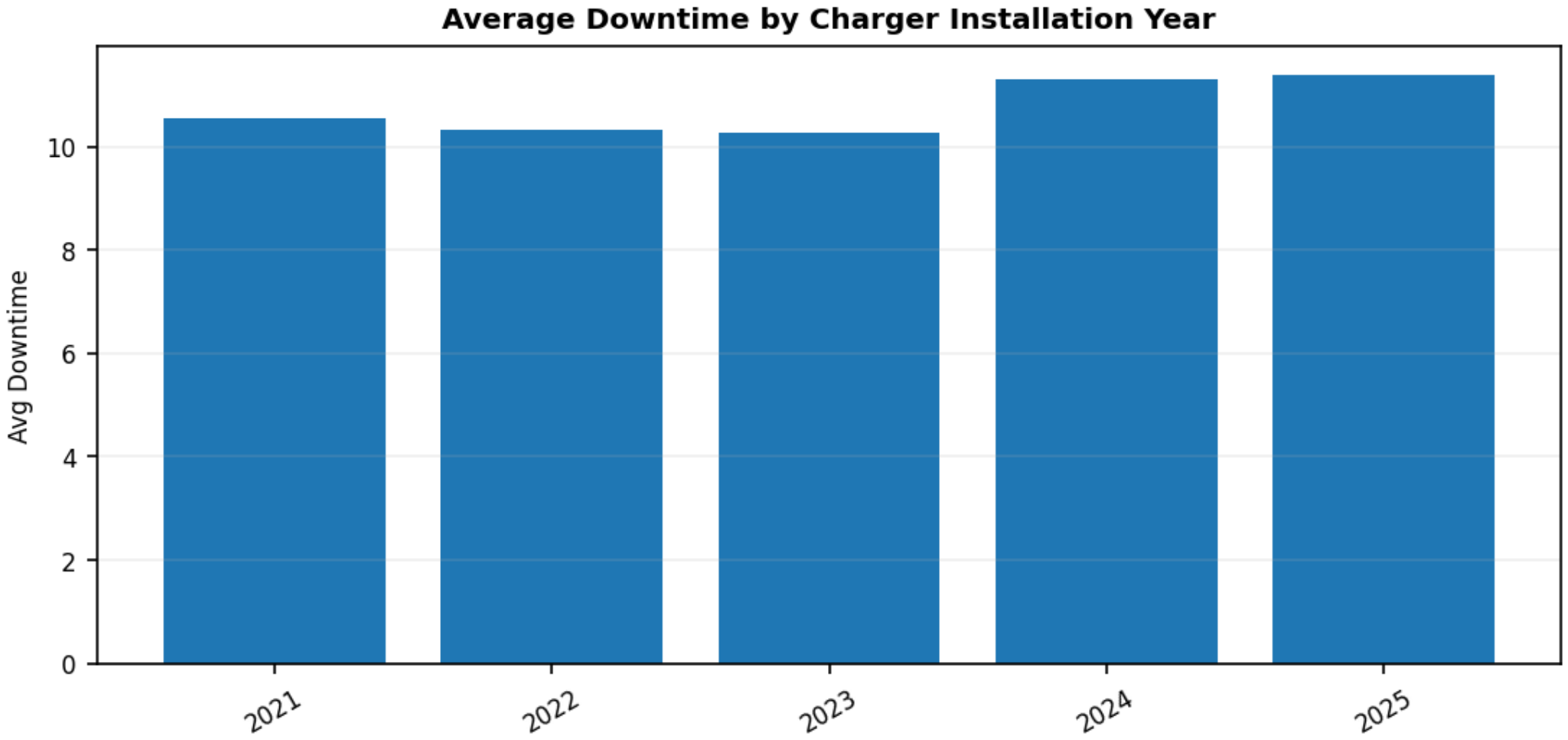
Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q11. Operator Performance Benchmarking



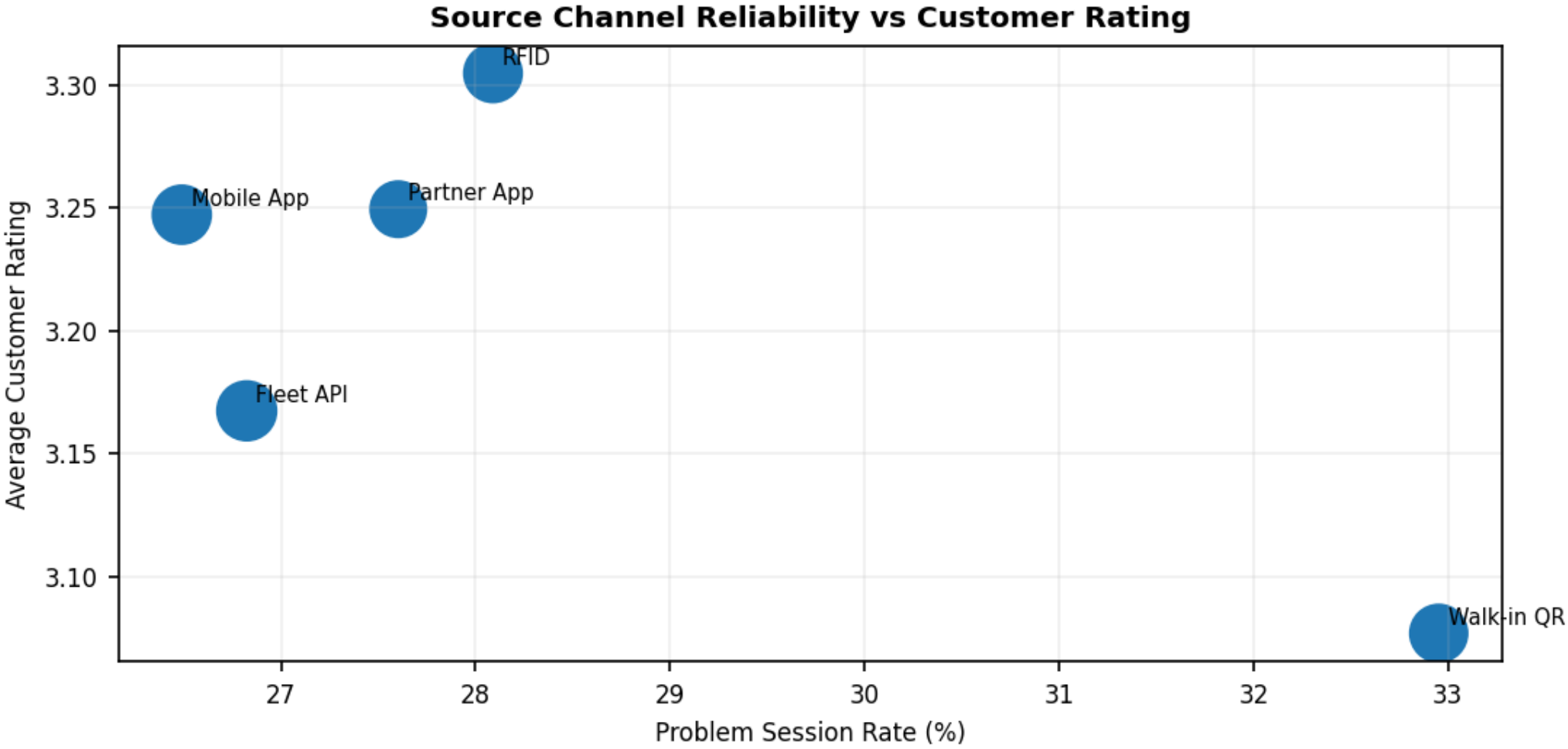
Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q12. Average Energy by Charger Type



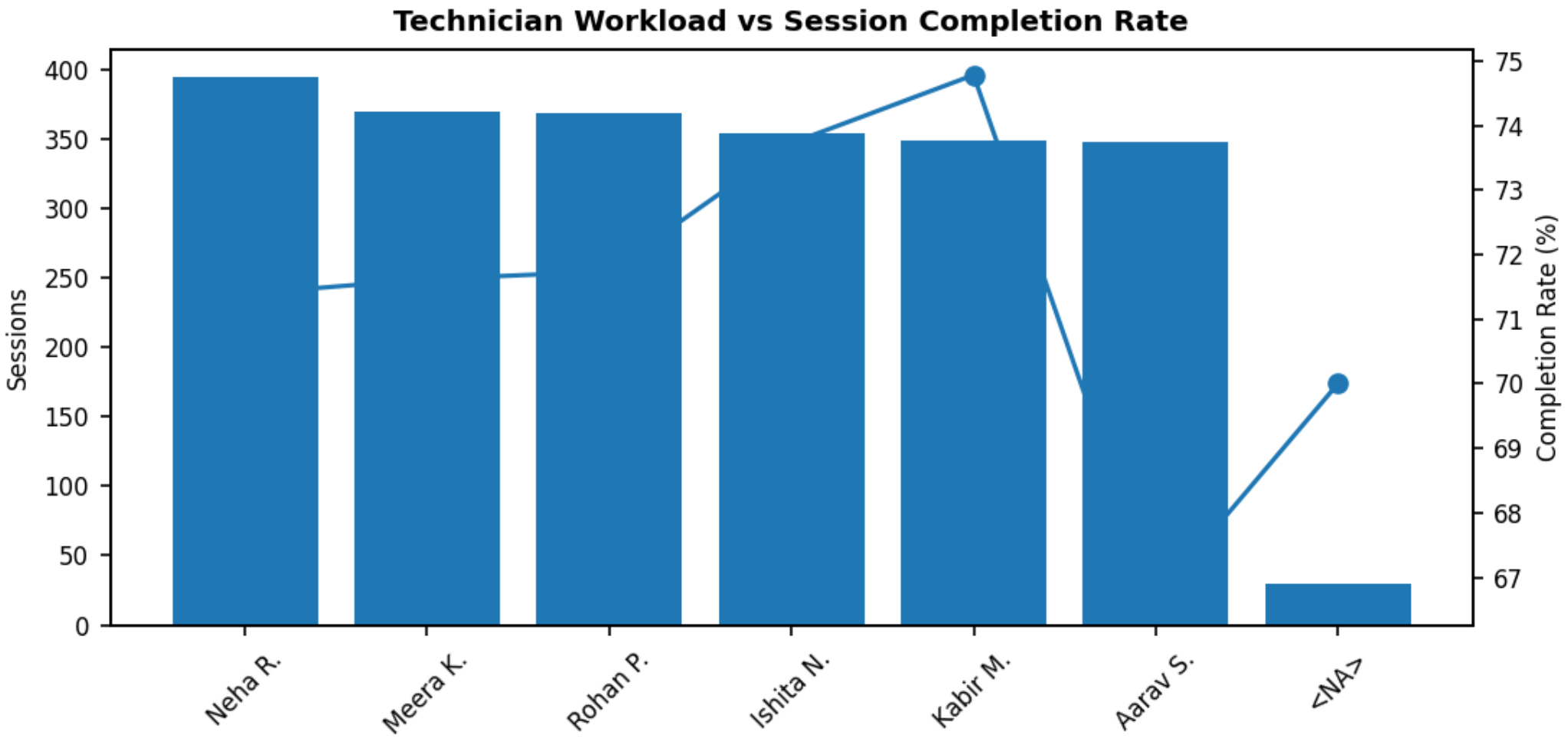
Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q13. Customer Ratings by Access Channel



Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q14. Technician Performance and Workload



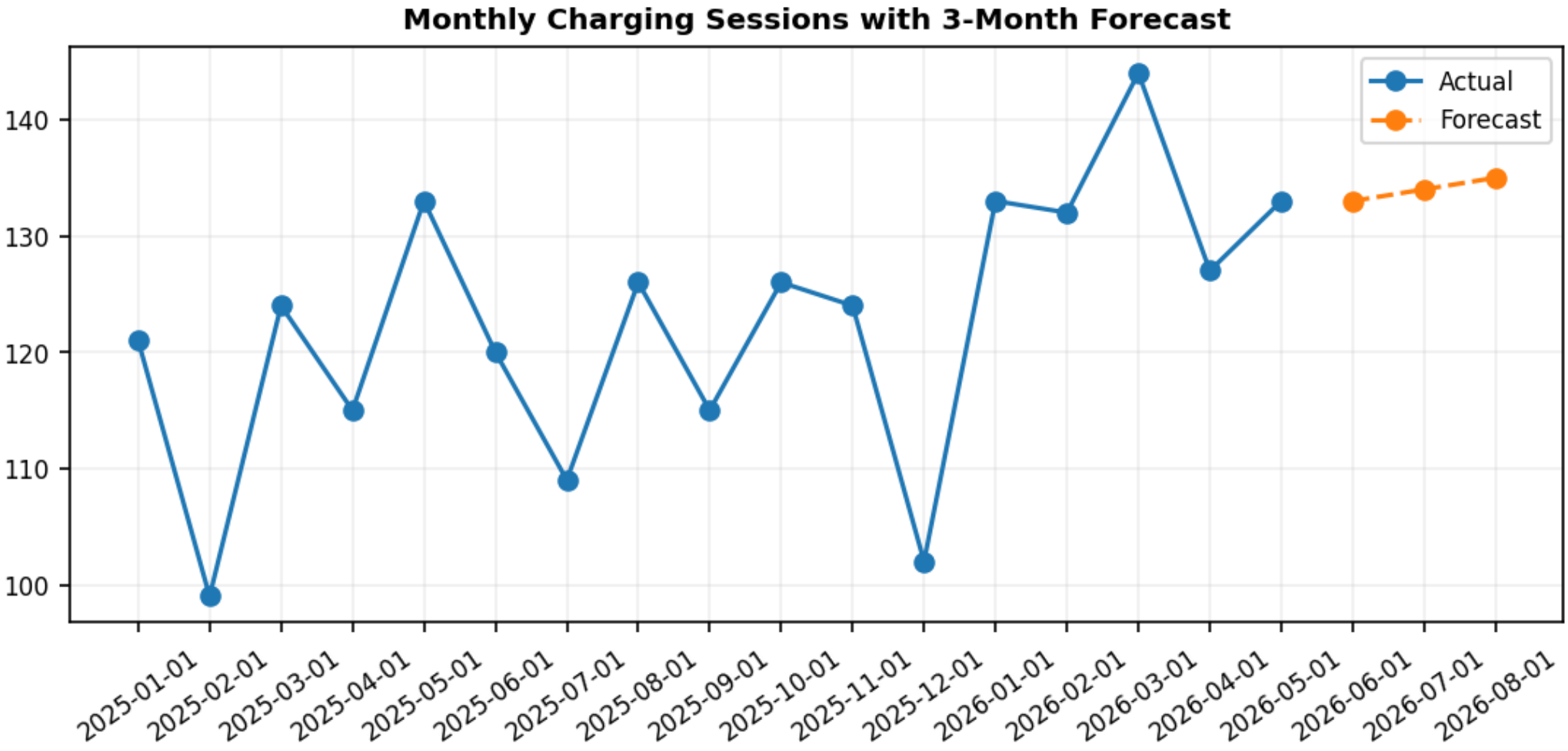
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Q15. Customer Experience Index by City



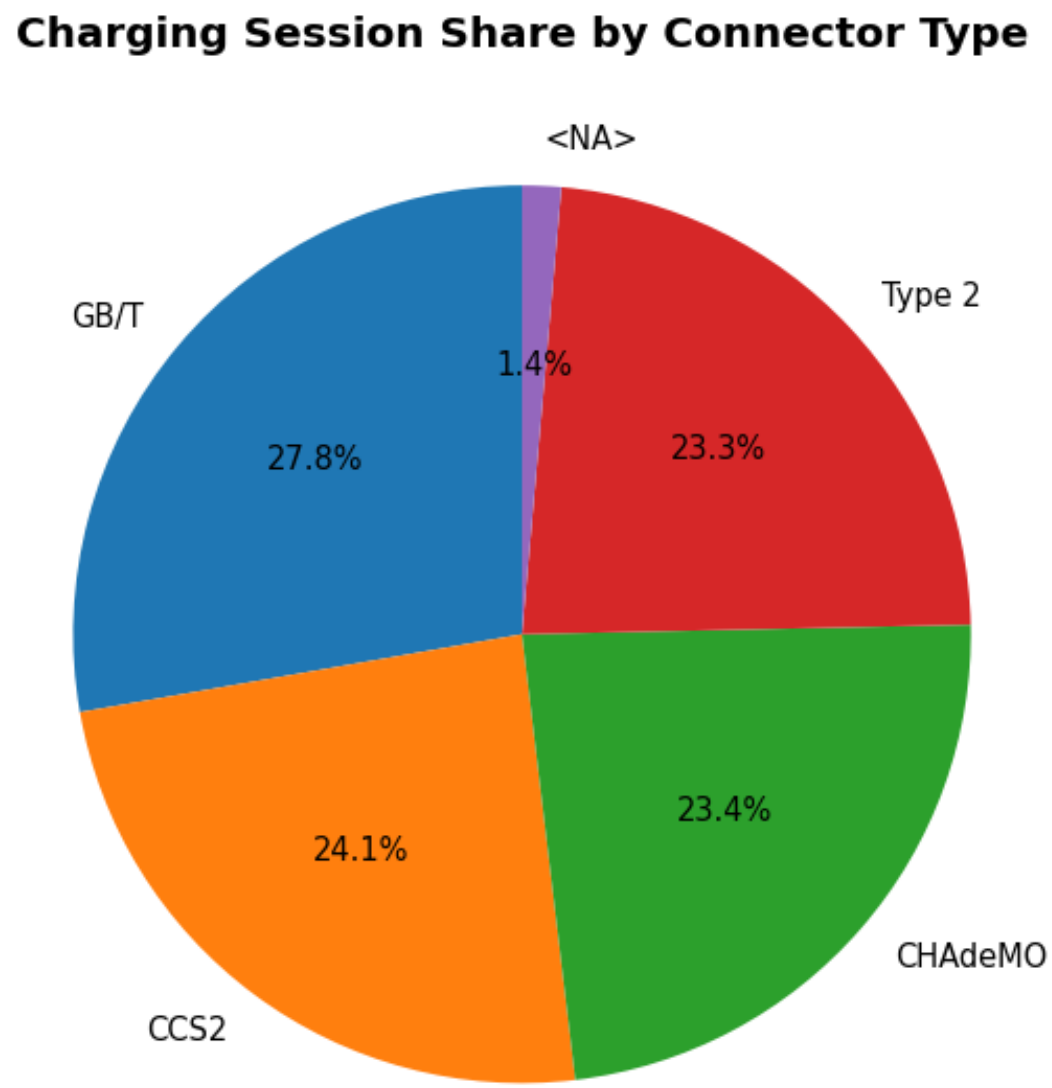
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Q16. Monthly Charging Demand Forecast



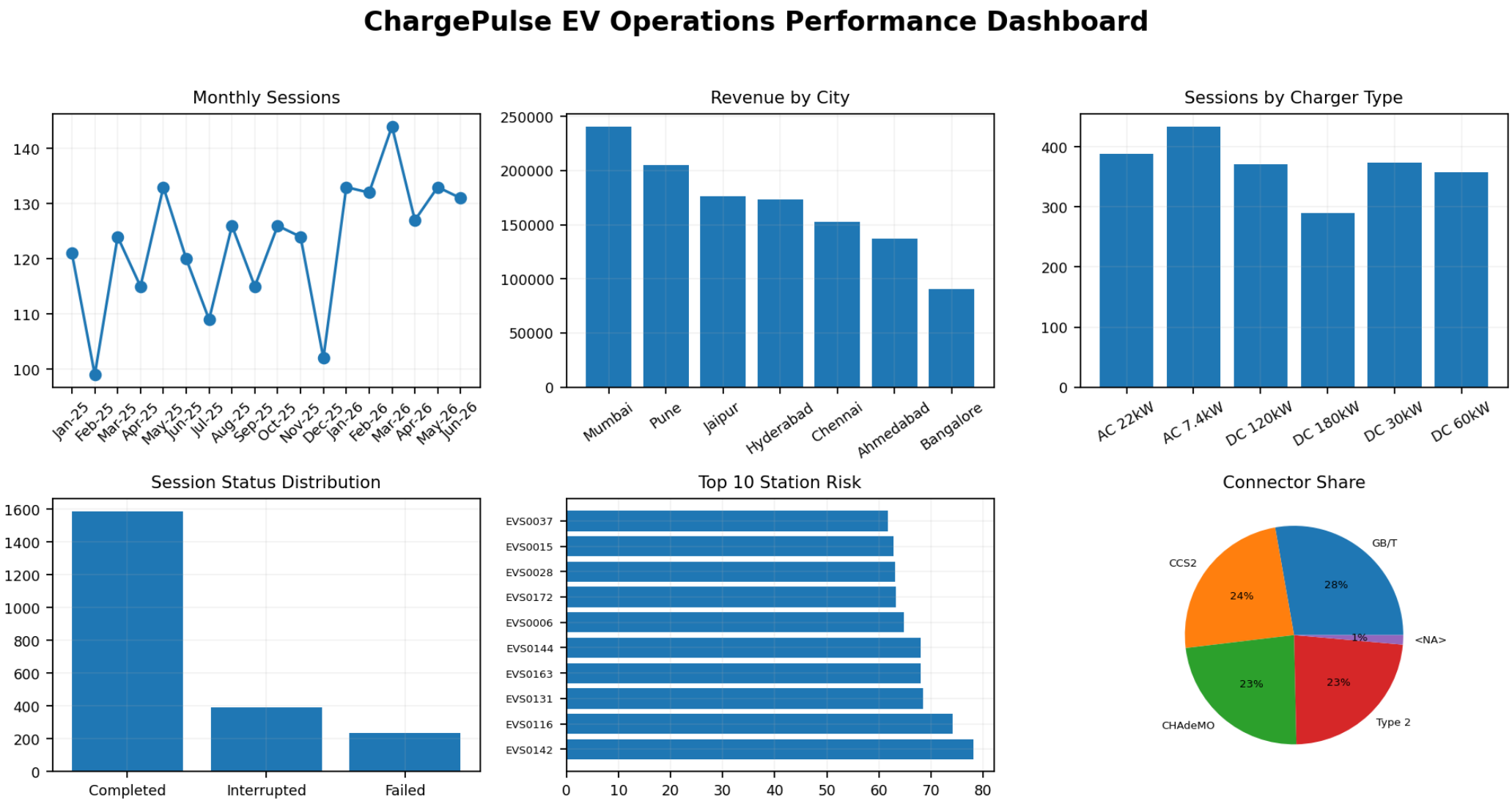
Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q17. EV Connector Demand Mix Analysis



Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.

Q18. ChargePulse EV Operations Performance Dashboard



Readable visualization copy: chart area enlarged to preserve axis labels, category names, and titles.